Journal of Financial Stability 75 (2024) 101336

Contents lists available at ScienceDirect

Journal of Financial Stability

journal homepage: www.elsevier.com/locate/jfstabil

Estimating the impact of supply chain network contagion on financial stability

Zlata Tabachová [a], Christian Diem [a,b], András Borsos [c,a], Csaba Burger [d], Stefan Thurner [a,e,f,*]

[a] Complexity Science Hub Vienna, Josefstädter Straße 39, A-1080, Austria
[b] Institute for Finance, Banking and Insurance, WU Vienna University of Economics and Business, Welthandelsplatz 1, A-1020, Austria
[c] Financial Systems Analysis Department, Central Bank of Hungary, Szabadság tér 9, Budapest 1054, Hungary
[d] Directorate Statistics, Central Bank of Hungary, Szabadság tér 9, Budapest 1054, Hungary
[e] Section for Science of Complex Systems, Medical University of Vienna, Spitalgasse 23, A-1090, Austria
[f] Santa Fe Institute, 1399 Hyde Park Road, Santa Fe, NM 87501, USA

ARTICLE INFO

Keywords:
Supply chain networks
Financial networks
Contagion
Stress testing
Systemic risk
Multi-layer networks

ABSTRACT

Realistic credit risk assessment, the estimation of losses due to a debtors failure, is central for maintaining financial stability. Credit risk models focus on the financial conditions of borrowers and only marginally consider other risks from the real economy, supply chains in particular. Recent pandemics, geopolitical instabilities, and natural disasters demonstrated that supply chain shocks can contribute to financial losses large enough to threaten financial stability. Based on a unique nation-wide micro-dataset, containing practically all supply chain relations of all Hungarian firms, together with their bank loans, we develop a multi-layer shock propagation framework to estimate how economic shocks to firms cascade in the supply chain network (SCN), leading to additional financial losses to firms, additional defaults of loans and, hence, losses to banks' equity buffers. First, we estimate the financial systemic risk of individual firms, by quantifying the expected financial losses caused by a firm's own- and all the secondary defaulting loans caused by supply chain network contagion. We find a small fraction of firms carrying substantial financial systemic risk, affecting up to 22% of the banking system's overall equity (assuming a loss given default of 100%). These losses are predominantly caused by SCN-contagion. Second, we calculate for every bank the expected loss (EL), value at risk (VaR) and expected shortfall (ES), with and without SCN-contagion. We find that SCN-contagion amplifies EL, VaR, and ES by a factor of 5.2, 6.7 and 4.4, respectively. Third, we showcase how the new framework can be used to assess the risks of a large real economy shock for financial stability. We simulate the financial losses from a COVID-19 inspired shock calibrated from firm-level employment data in the beginning of 2020. Our simulations show that without any interventions, system-wide bank equity would suffer losses of 6%. The framework can be used to design and test targeted policy interventions, e.g., optimally providing firms with enough liquidity-support to avert their default. By supporting *selected* illiquid (yet solvent) firms with additional liquidity totalling 0.5% of overall bank equity, the losses can be reduced from 6% to 1% of overall bank equity. These findings indicate that for a more complete picture of financial stability and realistic credit risk assessment, SCN contagion needs to be considered. This now quantifiable contagion channel is of relevance for future systemic risk assessments of regulators.

1. Introduction

Credit risk (CR) assessment is central for banks to operate in an economically sustainable way. CR materializes when a counterparty is unable or unwilling to fulfil its contractual debt obligations. In 1988 the *Basel Committee on Banking Supervision* (BCBS) published the first *Basel Accord* (BCBS, 1988) providing a set of minimum capital requirements that must be met by banks to guard against CR. These requirements have been refined also in the following accords *Basel II* (2004) (BCBS, 2004) and *Basel III* (2010) (BCBS, 2011), reflecting the importance of CR for financial stability. The regulatory path has been evolving towards a system where capital requirements are explicitly linked to the risk of activities undertaken by banks. This means that as perceived default risks increase, so do the risk-weights and the bank's capital buffer. Therefore, for being able to properly determine adequate capital

* Corresponding author at: Complexity Science Hub Vienna, Josefstädter Straße 39, A-1080, Austria.
E-mail addresses: tabachova@csh.ac.at (Z. Tabachová), diem@csh.ac.at (C. Diem), stefan.thurner@meduniwien.ac.at (S. Thurner).

https://doi.org/10.1016/j.jfs.2024.101336
Received 28 April 2023; Received in revised form 29 May 2024; Accepted 10 October 2024
Available online 18 October 2024

buffers and adjust their interest rates, it is essential for banks to have an in-depth knowledge of all sources of CR.

Recent crises like the COVID-19 pandemic, the conflict in Ukraine, and several natural disasters have impressively shown that the propagation of shocks along supply chain networks (SCNs) lead to large financial losses of firms, and as a consequence, also to their creditors if no government interventions take place. It was shown that shock propagation along supply chains can be dramatic, for example in the case of Hurricane "Katrina" (Hallegatte, 2008), the Japanese earthquake of 2011 (Carvalho et al., 2021; Inoue and Todo, 2019), natural disasters in the US (Barrot and Sauvagnat, 2016), the first lockdown during the COVID-19 crises in the UK (Pichler et al., 2022b; Diem et al., 2024a), disruptions of natural gas supply (Pichler et al., 2024), or failures of systemically important firms (Diem et al., 2022a). The COVID-19 crises led to substantial shock propagation effects along SCNs and affected almost every sector, including the especially vulnerable small and medium enterprises (SMEs) (Bartik et al., 2020). Banks only did not suffer extensive losses from non-performing loans during the pandemic by the virtue of enormous liquidity support measures, furlough schemes, and other measures imposed by governments and central banks (Milstein and Wessel, 2021; ECB, 2020). In the future, an accelerating climate crisis is likely to increase the frequency and magnitude of natural disasters, which will make the propagation of shocks along supply chains an even larger concern (Willner et al., 2018; Battiston et al., 2021).

To date, shock propagation through firm-level SCNs has not yet reached its ways into quantitative financial risk assessment of banks, or stress testing methodologies used by regulators; it has only recently been scarcely featured by research. Traditionally, credit risk models focus on the financial conditions of the borrowers, using their financial statements as the most relevant inputs. A rating assessment of borrowers usually involves variables derived from *balance sheets*, *income statements*, and *cash flow statements*. For example, according to the BCBS, the *leverage ratio* (*total assets* divided by *total equity*) has the strongest explanatory power, especially when combined with *revenue*, see BCBS (2001). Credit rating models that are applied on these variables, are primarily based on classical statistical methods, such as *logistic regression* (Cox, 1958).[1] Credit risk management evaluates information about suppliers and buyers in given markets or under specific economic conditions in, predominantly, qualitative ways (Gorgijevska and Gorgieva-Trajkovska, 2019; Moretto et al., 2019). The subjective analysis of clients' supply chain exposures can complement existing quantitative credit ratings, but is limited to first-tier supplier-buyer relations and, hence, cannot capture the complex structures of SCNs that transmit disruptions far beyond the first tier (Carvalho et al., 2021).

Research on network-based financial systemic risk (Boss et al., 2004; Battiston et al., 2012; Thurner and Poledna, 2013; Poledna et al., 2015, 2021; Diem et al., 2020; Feinstein et al., 2017; Gai and Kapadia, 2010; Thurner, 2022; Poledna et al., 2018), and macro prudential stress testing (Farmer et al., 2022; Cont and Schaanning, 2017; Glasserman and Young, 2016; Gauthier et al., 2012; Cont et al., 2010; Elsinger et al., 2006), primarily focuses on the nature of the propagation of shocks in *financial* networks, but does not include contagion effects along the SCNs, the back bone of the real economy. Further examples of this stream of literature include Buncic and Melecky (2013), Acharya et al. (2014), Levy-Carciente et al. (2015), Arnold et al. (2012), Borio et al. (2014), Vazquez et al. (2012), Bookstaber et al. (2016) and Gould et al. (2021), or Carro and Stupariu (2024). Early attempts of agent based approaches to linking the real economy to the financial system do not feature SCNs, but centralized product markets (Klimek et al., 2015). As pointed out in Battiston and Martinez-Jaramillo (2018), the literature connecting the real economy with the financial system is strongly under-researched. Herring and Schuermann (2022) and Potter and Schuermann (2020) emphasize that the framework of current stress testing should be broadened to include non-financial threats to financial stability, such as economic or climate shocks. Similarly, Farmer et al. (2022) highlights the need for so-called *system-wide stress testing* to account for relations between economic and financial systems. Several regulating bodies and institutions such as the BCBS BCBS (2021a), BCBS (2021b)and BCBS (2022), the Federal Reserve (FED) (Brunetti et al., 2021), and the European Central Bank (ECB) (ECB/ESRB, 2022) pointed out the significance of climate-related risk to financial stability and identified supply chains as one of the relevant risk transmission channels.

Due to a lack of granular data, so far it has been impossible to quantify exposures of financial systems to contagion in SCNs on the firm-level. On the industry level contributions in this direction were provided by Guth et al. (2020) where aggregated supply chain networks in the form of IO tables were used to assess the impact of supply chain disruptions in the context of the COVID-19 crises on the Austrian banking system. However, the use of aggregated industry-level production network data can cause substantial mis-estimations of production losses (Diem et al., 2024a), showing the need for firm-level modelling approaches of supply chain shock propagation. Recently large-scale firm-level supply network data sets have become available for a dozen of countries (Bacilieri et al., 2022; Pichler et al., 2023), yet there is only a handful of papers focusing on the interaction of firm-level SCNs with the financial system. The spreading of liquidity shortages to the production network (Huremovic et al., 2020; Demir et al., 2022) was explored, which can also generate feedback effects for the financial sector (Silva et al., 2018). However, the first works considering explicit interactions between the firm-level production network and the financial sector using granular network data for both systems are Huremovic et al. (2020) and Borsos and Mero (2020), are limited to shocks that originate in the financial sector.

Here we present a data-driven computational framework for estimating how the initial failure of firms spreads along the supply chain network (SCN) within a country, leading to potentially additional firm defaults. These spread to the financial system through additional firm-loan write-offs and equity losses for banks. Based on a nation-wide micro-dataset, containing all supply chain links of all Hungarian firms in combination with comprehensive credit registry data containing all commercial loans that firms obtained from banks, we are in the unique position to address the following two research questions. First, to what extent does systemic risk in the real economy – created by cascades of production failures in national SCNs – affect financial stability? And second, how is the credit risk exposure of banks — measured by expected loss, value at risk, and expected shortfall – amplified by contagion in these SCNs, or, equivalently, how much is credit risk underestimated by ignoring supply chain contagion? To answer these questions, we take two perspectives, a system-wide and one that is bank-specific. First, we introduce for every firm a *financial systemic risk index* (FSRI). FSRI quantifies the financial losses of the banking system, caused directly by the firm's failure and indirectly from the resulting propagation of shocks in the supply chain network. Second, we stress the system with 10,000 different initial shock scenarios (each shock is an iid draw from firms' empirically estimated probability of defaults)

[1] By now, machine learning and deep learning based credit risk models outperform classical ones in terms of accuracy (Shi et al., 2022). They can handle large alternative data sets, such as text- or graph data (Hamilton, 2020) including the extraction of relevant information from these. These approaches could be used to include SCN data. As discussed in the *European Banking Authority report on big data and advanced analytics* (EBA, 2020), there is an evident tendency in employing *big data and advanced analytics (BD&AA)* into many aspects of the banking business, such as fraud detection or client interactions. There is also growing interest in the area of financial risk management. On the other side, there are still many concerns like "biasness" or the fact that (BD&AA) methods still largely remain "black boxes". That is why regulators so far have been cautious of approving their usage for risk management.

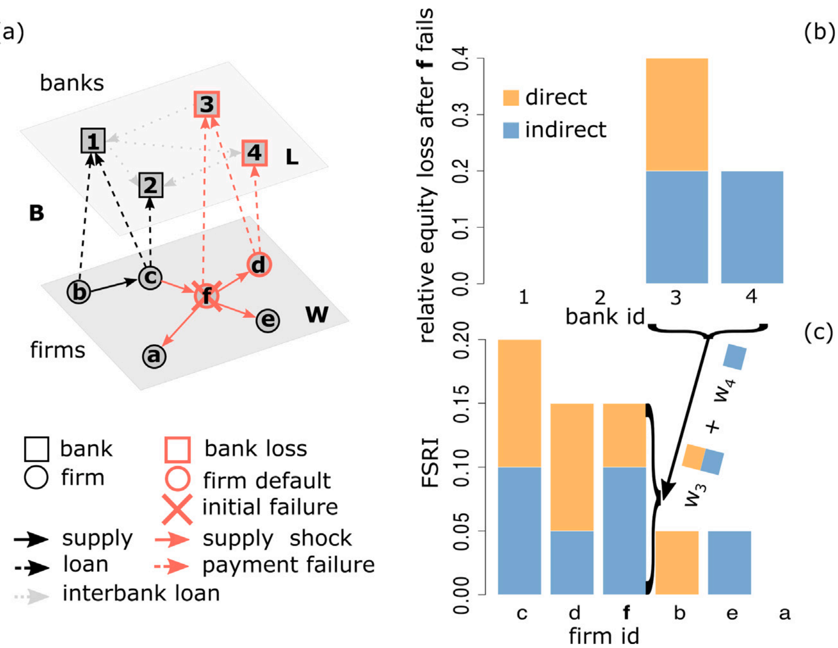

Fig. 1. Schematic view of contagion in the firm-bank multi-layer network. (a) The bottom layer, W, represents the supply chain network. Every node is a firm (circle) and a link, W_{ij}, (solid arrow) represent a supplier-buyer relation, from supplier i to buyer j. The top layer represents the inter-bank network, L. Nodes are banks (square) and a link, $L_{k\ell}$, (dotted arrow) is a liability bank k has to bank ℓ. Inter-bank links are not considered here, $L_{k\ell} = 0$. Layer L is connected with the supply chain network, W, through bank-firm loans given by the matrix, B, (dashed arrows). B_{ik} is a liability of firm i to bank k. In this example, we show how contagion starts with the initial failure of firm f (red X) (assuming a 100% production disruption), and spreads upstream to firm c and downstream to firms a, d, and e (full red arrows). As a result, production levels of those firms are reduced, leading to financial losses for firms f and d, that cause their default (red circles). Whenever a firm defaults, banks write off the corresponding loans in their balance sheets, resulting in losses of equity and (possibly) liquidity in the financial layer. (b) losses from the default of f and d are presented as bar-plot, with bank indices on the x-axis and the relative equity losses of the involved banks on the y-axis. In the case of the initial failure of f, bank 4 suffers a 20% loss of equity *indirectly* (blue bar). Similarly, bank 3 suffers a 20% loss of equity directly from the loan to firm f, and an additional 20% indirect loss from the additional default of firm d due to supply chain contagion. (c) Overall losses in the bank layer are shown as a bar-plot with firm indices on the x-axis and financial systemic risk index (FSRI) on the y-axis. FSRI is the weighted average equity loss of all banks, with the weights given by the relative equity size of each bank, i.e. $w_3 = w_4 = 0.25$, since we assume that all banks in this example have the same equity; see main text. For the third bar we see, that FSRI of firm f is equal to 15%, meaning that 15% of total equity in the financial sector can be lost if f fails. Analogically, firms c, d, b, e, and a can cause 20%, 15%, 5%, 5%, and 0% of damage, respectively. Financial contagion between banks is not considered. In the same way we can assess more general shocks that affect multiple firms simultaneously. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

and estimate additional risks for every single bank by generating loss distributions that take SCN contagion into account. In particular, we compute the *expected loss* (EL), the *value at risk* (VaR), and the *expected shortfall* (ES) for all banks with and without SCN contagion. Third, we showcase how the shock propagation framework can be used to assess the risks of a large real economy shock for financial stability. We simulate the SCN- and financial contagion of a COVID-19 inspired shock calibrated from firm-level employment data in the beginning of 2020 and design a targeted policy intervention that optimally provides firms with enough liquidity support to avert their default. In this way we extend the economic systemic risk index (ESRI) (Diem et al., 2022a) – that quantifies the supply-network-wide production (output) losses caused by the initial failure of a single firm or group of firms – to account for financial losses and potential contagion effects to the financial system.

2. Model and methods

To describe how the propagation of production disruptions along supply chains spreads to banks, we use the supply chain network (SCN) of n firms given by the $n \times n$ matrix, W. Its elements, W_{ij}, are the sales volumes of firm i to firm j, or equivalently, purchases of firm j from i in monetary units (Euros per year). The second network layer is the interbank network, L, containing m banks, where a link, L_{ij}, represents a liability of bank i towards bank j. Note that L_{ij} could also represent any other direct exposure, such as studied in Poledna et al. (2015). Every bank k is endowed with equity, e_k. The two network layers are connected through the $n \times m$ firm-bank loan matrix, B, where B_{ik} is the *outstanding amount* of the liability firm i has towards bank k. In this paper we focus on contagion from the supply chain, W, to the interbank network, L. We omit potential cascading effects inside L.[2] This model is calibrated with actual data derived from Hungarian VAT, balance sheet, and credit registry data for the year 2019 for W, L, B, and e. For details, see Section 3.

Schematically we describe the two-layer network in Fig. 1(a). The bottom layer represents the SCN, W, consisting of 6 firms (circles). Arrows between them indicate supplier-buyer relations (from supplier to buyer), e.g. b is a supplier of c and W_{bc} are the sales from b to c in Euros per year. The upper layer represents a financial network consisting of 4 banks (squares). Layers are connected by firm-bank loans, B (dashed arrows), e.g. firm f has a loan from bank 3, with B_{f3} being the outstanding amount. All firm loans are assumed to be of size $B_{ik} = 1$ and all interlayer links are, $L_{k\ell} = 0$ (light grey; no contagion between banks is possible). The capital or equity, e_k of all banks is set to 5. Contagion spreading from the SCN layer, W, to the individual banks occurs in three steps:

1. quantify the propagation of production disruptions in the SCN after a firm's initial failure;
2. update equity and liquidity buffers of firms in response to potentially reduced production levels and check if they became illiquid or insolvent and, hence, default on their loans;
3. update the banks' equity buffers by writing off the loans of the initially failed (and defaulted) firms, and the firms that defaulted due to supply chain network contagion.

[2] Interbank financial contagion channels could be straight forwardly included as an extension of the model presented in this work.

In this multi-layer network micro-simulation model, the actual data for W, L, B, and e is considered to be the realization of an "undisrupted" economy, where firms normally buy and sell in the SCN. The data contains firms' income statements, balance sheets and the outstanding principles of banks' commercial loan portfolios of the respective year. Given this initial state of the economy, we can next simulate the effects on this economy when it is affected by an initial shock that affects the production capacities of firms and propagates through the supply chain network. This will now lead to counterfactual realizations of income statements and balance sheets of the firms. From these we infer which firms would have defaulted on their loans if the initial shock had happened. Banks write off the respective loans and suffer hypothetical losses, given the initial firm shock. We next describe the three steps of the model in more detail.

Step 1. To simulate how a specific firm failure spreads through the SCN, W, we use the firm-level shock propagation mechanism of Diem et al. (2022a), described in Appendix A. There, a *Generalized Leontief Production Function* (GLPF) is specifically calibrated for every firm, i, to determine how much it can still produce if one of its suppliers, j, fails to deliver its products. We refer to this as a supply shock or downstream shock propagation (from supplier to buyer). If a customer, ℓ, of firm i stops buying it faces a demand shock or upstream shock propagation (from buyer to supplier). The undisturbed production state of the SCN at time, t_0, is represented by the production-level vector, $h(t_0)$. Initially, each firm, i, produces 100% of its original production level, i.e. $h_i(t_0) = 1$ for every $i \in \{1, 2, \dots, n\}$. At time, t_1, an initial event occurs that disrupts the production of at least one firm in the SCN. This can be for example the (temporary) failure of a single firm, or a simultaneous production stop of many firms due to a COVID-19 lockdown or a natural disaster. The vector $\psi \in [0,1]^n$ denotes the severity of the initial shock by specifying the remaining production level for every firm after the initial event occurred. For instance, $\psi_i = 0.6$ means that firm i can still produce 60% of its pre-shock production level. We set the production level of every firm, i, to $h_i(t_1) = \psi_i$ ($h(t_1) = \psi$).[3] The initial shock, ψ, triggers an upstream and downstream shock propagation cascade along the supply chain links of firms. The propagation continues until every firm stops to adjust its production levels, i.e. $h_i(t_\tau) - h_i(t_{\tau+1}) < \epsilon$, and the system converges to a new stationary state at time $T := t_{\tau+1}$. This state is represented by the final remaining production levels, $h(T) \in [0,1]^n$, after the initial event. For example, $h_i(T) = 0.3$ means that firm i lost 70% of its original production level. We assume that production levels remain at this new state for a certain amount of time before firms can recover by finding new suppliers and buyers. Here, we do not model this rewiring of the network since realistic rewiring dynamics needs substantially more assumptions and data that would go beyond the scope of this work. Therefore, the model outputs represent the short term effects before firms can adjust their supplier and buyer structures.

In the example of Fig. 1(a) the initial shock scenario assumes the failure of firm f (red X), e.g. due to an exogenous event such as a fire. Initially, we have $h(t_0) = (1,1,1,1,1,1)$ corresponding to a 100% production level of all 6 firms a, b, c, d, e, and f, respectively. Next, the initial shock at time t_1 is introduced with the vector, $\psi = h(t_1) = (1,1,1,1,1,0)$, where the 6th element indicates the drop of f's production level to 0%. The shock now spreads (red arrows) upstream to f's supplier c and downstream to its buyers a, e, and d. To keep the example short, we assume here that the production of a, c, and e does not critically depend on the inputs of f, but d cannot produce without them, i.e. suffers a 100% production disruption. Consequently, after convergence at, T, the final production level vector is $h(T) = (1,1,1,1,0,0)$.

Table 1
Stylized balance sheet and income statement of firm, i.

Assets	Liabilities	Income Statement
Short term assets (a_i)	Accounts payable (s_i)	+ Revenues (r_i)
Inventory	Other liabilities	− Material costs (c_i)
Fixed assets	Equity (z_i)	Non-operational profit (o_i)
Other assets	Profit (p_i)	+ Other income
	Other equity	− Other costs
Total assets	**Total liabilities**	**Net profit** (p_i)

Step 2. The now reduced production levels, $h(T)$, cause a drop in firms' revenues and material costs that affects their profits. This change is relative to the pre-shock profits recorded in the income statement. Reduced profits may turn into large losses and potentially will affect the equity and liquidity of firms, and eventually, the ability to repay their bank loans (if they have any). To find out whether a company can withstand the financial losses from a given production disruption we use the information from its income statement and balance sheet, a stylized version of which is depicted in Table 1.

For every firm, i, we infer for a given year its *revenue*, r_i, its *material costs*, c_i, its *non-operational profit*, o_i, consisting of *other income* (e.g., income from financial assets, sales of land, etc.), and *other costs* (e.g. HR costs, depreciation, rent, and other costs that cannot be adjusted within a short period of time), from its income statement data. From the balance sheet data we read off firm i's *equity*, z_i, its *short-term assets*, a_i (including cash, short-term claims or accounts receivable, and securities), and its *short-term liabilities* (accounts payable) from buying goods and services from suppliers, s_i.

We next determine how a production reduction of firm, i, translates into a reduction of its revenues, r_i, and its material costs, c_i, affecting the profit, p_i. We assume that the non-operational profit stays the same, and that the revenue is reduced to $h_i(T)r_i$. Note, that for simplicity we neglect potential inventories of inputs and finished goods. We further assume that firms can reduce their material costs proportionally to the production level, i.e. material costs after the cascade are $h_i(T)c_i$. This assumes that firms only have to buy exactly the amount they need for production and neglects the fact that supply contracts might be long-term. This assumption might be overly optimistic. Since, the net profit is the sum of revenue, negative costs, and non-operational profit, $p_i = r_i - c_i + o_i$, the new profit at the reduced production level is $\tilde{p}_i = h_i(T)(r_i - c_i) + o_i$ and the change in profit is

$$\Delta p_i \left(h_i(T)\right) = p_i - \tilde{p}_i = \left(1 - h_i(T)\right)(r_i - c_i) \quad . \tag{1}$$

The change in profits affects a firm's equity and its liquid assets (via the cash flow statement). The equity, z_i, of firm i includes the equity from the previous year and the net profit of the current year, p_i. Note, that p_i can also be negative. After accounting for the change in profits the new equity is $\tilde{z}_i = z_i - \Delta p_i$. According to The Insolvency Code (Act, 1991) (Section 27 (2f)) firm i becomes insolvent if $\tilde{z}_i < 0$. The cash position of the firm at the balance sheet reporting date depends on the profits, p_i, made during this period.[4] The cash flow statement structure implies that a reduction of profits will also cause a decrease of the firm's

[3] Note that initial shocks could affect firms at different time points, i.e. $\psi_i(t_i)$ happens before $\psi_j(t_j)$ if $t_i < t_j$. Here we always assume that $t_i = t_j = t_1 \ \forall \ ij$ and omit the time index in the notation. The model can be easily adjusted to feature different time points of disruption events. In the presented model the time intervals between two elements of the sequence $t_1, t_2, t_3, \dots, T$ correspond to the same time unit, e.g., a calendar day or a calendar week.

[4] The explicit link between profits and cash is the cash flow statement. The indirect method for calculating the cash flow during the year starts with the profit p_i and adds or subtracts transactions that are not cash effective and eventually yields the change in cash position. Examples for non cash effective transactions are adding back depreciation, adding a reduction of accounts receivable, deducting an increase in inventory levels, or deducting the purchase price of a new machine.

cash position. If cash turns negative the firm becomes illiquid, i.e., it can no longer pay its bills or repay loans, and may become bankrupt according to the Insolvency Code (Act, 1991) (Section 27 (2a)). We define the buffer against illiquidity more broadly than cash by using the sum of short-term assets (that could be sold fast) minus the short-term liabilities from goods and services (that are due soon), i.e. $a_i - s_i$.[5] After the shock, firm i has a liquidity of $a_i - s_i - \Delta p_i$ and if this is negative we assume that firm i declares bankruptcy, defaults on its loan, and banks write off all loans extended to i. Due to the cash flow calculation procedure and the assumption that the other transactions happen in the same way as in the case where the shock did not occur, the change in liquidity is Δp_i. This assumption presumably leads to an over-estimation of the liquidity loss as firms in distress would possibly defer investments into new machinery that are cash flow negative or could try to receive additional loans from banks (cash flow positive) to avoid becoming illiquid. We abstain from modelling this behaviour in detail as it involves a number of additional assumptions that cannot be backed with the available data. Instead in Appendix C we provide a sensitivity analysis for the size of liquidity shocks firms can withstand by assuming that firms can generate up to 100% additional liquidity. Note that for most firms the liquidity buffer is smaller than the equity buffer, hence, in our framework defaults from illiquidity will play a larger role than the ones caused by insolvency.

To sum up, we derived two insolvency conditions for firms that depend on the size of the firm's production reductions, $1-h_i(T)$, and its financial strength (liquidity, $a_i - s_i$, and equity buffers, z_i). We assume that either illiquidity or insolvency of a firm leads to its bankruptcy, which reflects the unlikeliness of loans to be repaid. The *unlikeliness to pay indicator* is one of the conditions under which a bank classifies its client as defaulted, see, e.g., Article 178(1-3) of Regulation (EU) No 575/2013 (RegulationEU, 2013). To conclude step 2, for all firms we define a binary default indicator

$$\chi_i = \begin{cases} 1 & \text{if} \quad (z_i - \Delta p_i) \le 0 \quad \text{or} \quad (a_i - s_i - \Delta p_i) \le 0 \quad , \\ 0 & \text{if} \quad (z_i - \Delta p_i) > 0 \quad \text{and} \quad (a_i - s_i - \Delta p_i) > 0 \quad , \end{cases} \tag{2}$$

where $\Delta p_i = \Delta p_i\left(h_i(T)\right)$, see Eq. (1). It indicates, which firms defaulted ($\chi_i = 1$) as a result of an initial shock, ψ, and the ensuing supply chain contagion after the cascade stops at time T. $\chi_i = 0$ means i is still a going concern (not defaulted). To see the role of the equity- and liquidity default conditions in Eq. (2) individually, we define the equity default indicator vector, χ_{eq}, that only considers the condition $(z_i - \Delta p_i) \le 0$, and the liquidity default indicator vector, χ_{l} that only checks the condition $(a_i - s_i - \Delta p_i) \le 0$. Note, that the union of the two vectors gives χ. In the following we define all equations in terms of χ, however, all definitions can be rewritten in terms of χ_{eq} and χ_{l} to analyse the effects of liquidity or equity defaults individually.

Step 3. Loans of defaulted firms are classified as *non-performing loans* (NPL). We assume that borrowers are not subject to any forbearance solutions (as would be standard procedure, see BCBS, 2017), and they cause write-offs or provisions for banks that reduce the banks' equity buffers. Impairment allowances for non-performing loans (or *loan loss provisions*) implied by current accounting practice are described in the International Financial Reporting Standards (IFSR 9) (IASB, 2014) or in the *guidance on credit risk and accounting for expected credit losses* (BCBS, 2015). For simplicity, we assume a short-term loss-given-default (LGD) of 100% across all firms, i.e., the entire exposure, B_{ik}, bank k has towards each firm i at the time of firm's default is written off. In practice long-run LGDs are substantially lower and the standard value set by the Basel Committee on Banking Supervision (2023) is 40%. We indicate the LGD with the variable $\kappa \in [0,1]$, which we set, $\kappa_{ik} = \kappa = 1$, for any i and k in the main text. We provide a sensitivity analysis of our results wrt. κ in Appendix B. We distinguish between *direct* and *indirect* losses of banks' equity. *Direct losses* are caused when a debtor, i, immediately defaults in response to the initial disruption, i.e., $0 \le \psi_i < 1$ and $\chi_i = 1$.

Accordingly, we define the *direct default indicator* via binary n-dimensional vector $\chi^{\text{dir}} = \chi^{\text{dir}}(\psi)$. *Indirect* bank losses are caused when a firm i defaults, $\chi_i = 1$, but not due to its initial production disruption, i.e. $\psi_i = 1$, but because of the additional production losses stemming from the propagation of the initial disruption. These equity losses of banks would not occur if supply chain network contagion is not present. The *indirect default* indicator $\chi^{\text{indir}} = \chi^{\text{indir}}(\psi, h_i(T))$ is defined as the binary n-dimensional vector

$$\chi_i^{\text{indir}} = \begin{cases} 1 & \text{if } \chi_i\left(h_i(T)\right) = 1 \text{ and } \chi_i\left(h_i(t_1)\right) = 0 \quad , \\ 0 & \text{if } \chi_i\left(h_i(t_1)\right) = 1 \quad . \end{cases} \tag{3}$$

Then, $\chi_i^{\text{dir}} = \chi_i - \chi_i^{\text{indir}}$ holds.[6] The fraction of lost equity, $\mathcal{L}_k$, of bank, k, — after the initial shock, ψ, has propagated through the supply chain network —, is calculated as

$$\mathcal{L}_k \equiv \sum_{j=1}^{n} \chi_j \frac{\kappa_{jk} B_{jk}}{e_k} = \sum_{j=1}^{n} \left(\chi_j^{\text{dir}} + \chi_j^{\text{indir}}\right) \frac{\kappa_{jk} B_{jk}}{e_k} \quad , \tag{4}$$

where $\mathcal{L}_k = \mathcal{L}_k(\chi(\psi), \kappa)$ is calculated for every bank $k \in \{1, 2, \dots, m\}$, and κ_{jk} is the loss-given-default for the respective loan, B_{jk}. Since, the elements of B are given in monetary units, we divide the loans, B_{jk}, by the respective bank's equity, e_k. The fraction of the lost equity from direct defaults is $\mathcal{L}_k^{\text{dir}} = \sum_{j=1}^{n} \chi_j^{\text{dir}} \frac{\kappa_{jk} B_{jk}}{e_k}$ and from the supply chain contagion induced defaults $\mathcal{L}_k^{\text{indir}} = \sum_{j=1}^{n} \chi_j^{\text{indir}} \frac{\kappa_{jk} B_{jk}}{e_k}$. The equity losses suffered by the entire banking sector from initial shock, ψ, is

$$\mathcal{L}(\psi, \kappa) \equiv \frac{\sum_{k=1}^{m} \mathcal{L}_k(\psi, \kappa) e_k}{\sum_{\ell=1}^{m} e_\ell} \quad . \tag{5}$$

The financial loss of an individual bank is measured as a fraction of its equity and defined in Eq. (4) in short we write *bank equity losses*. The system wide financial losses are measured as a fraction of the overall equity across all banks and defined in Eq. (5), in short we write *financial system losses*.

To illustrate steps 2 and 3 in Fig. 1(a) the red circles indicate that firms f and d defaulted, i.e $\chi = (0,0,0,1,0,1)$. We consider that firms a, b, c, and e were able to withstand the shock, $\psi = (1,1,1,1,1,0)$ caused by the initial disruption of f. Red dotted arrows indicate payment failures of defaulted firms f and d, red squares show that banks 3 and 4 suffered losses. Bank 3 suffered losses from two non-performing loans, B_{f3} (directly) and B_{d3} (indirectly). Bank 4 suffered losses only indirectly from B_{d4}. For this scenario, $\chi^{\text{indir}}(\psi) = (0,0,0,1,0,0)$ and $\chi^{\text{dir}}(\psi) = (0,0,0,0,0,1)$, $\chi = \chi^{\text{indir}} + \chi^{\text{dir}}$. Recall, in this example all banks have the same initial amount of equity of 5, and all loans have the same outstanding amounts of 1 an LGD of 100%. Relative losses of banks after the initial failure of firm f are depicted in Fig. 1(b), where every bar is associated to one bank. Banks 1 and 2 did not experience any losses, explaining why $\mathcal{L}_1(\psi) = \mathcal{L}_2(\psi) = 0$. Bank 3 lost 1/5 of equity directly (orange bar) and 1/5 of its equity indirectly (blue bar), i.e. $\mathcal{L}_3(\psi) = \mathcal{L}_3^{\text{dir}} + \mathcal{L}_3^{\text{indir}} = 2/5$. Similarly, we get $\mathcal{L}_4 = \mathcal{L}_4^{\text{indir}}(\psi) = 0.2$ and $\mathcal{L}(\psi) = (3/20) = 0.15$.

Estimating the effects of individual-firm failures on overall bank equity. We measure if an individual firm can affect financial stability through having high systemic risk in the supply chain network. We simulate the impact on the banking system (sum of all the banks' equities) caused by the full production disruption of a single firm, j, by computing the resulting supply chain contagion and loan defaults. For the initial failure of firm j we define the single-firm shock vector

$$\psi_i^{\text{S},j} = \begin{cases} 1 & \text{if} \quad i \ne j \\ 0 & \text{if} \quad i = j \quad . \end{cases} \tag{6}$$

[5] Note, we do not subtract short term liabilities to banks, as banks can introduce forbearance measures or provide additional liquidity to firms.

[6] Note, that to obtain χ^{dir} we do not need to perform contagion at all, meaning that Step 1 can be omitted, and in Step 2 we put $h(T) = \psi$. χ^{dir} is obtained accordingly from Eq. (2).

The value $\psi_i^{S,j} \in \{0,1\}$ indicates the production level of firm i under the initial shock scenario, j (failure of firm j).[7] The set $\Psi^{\mathrm{S}} = \{\psi^{S,1}, \psi^{S,2}, \dots, \psi^{S,j}, \dots, \psi^{S,n}\}$, collects the n possible shock scenarios, corresponding to a failure of a single firm. Note that we use the indices $i, j \in \{1,2,\dots,n\}$ to indicate firms in the SCN and $k \in \{1,2,\dots,m\}$ for banks. For every initial single firm failure, $\psi^{\mathrm{S},j}$, we perform *steps* $1-3$ and receive the fraction of lost equity, $\mathcal{L}_k = \mathcal{L}_k(\psi^{\mathrm{S},j})$, for every bank k. Based on the equity losses, $\mathcal{L}_k$, we compute the *financial systemic risk index (FSRI)* of firm j, as

$$\mathrm{FSRI}_j \equiv \sum_{k=1}^{m} \frac{e_k}{\sum_{\ell=1}^{m} e_\ell} \min\left[1, \mathcal{L}_k\left(\psi^{\mathrm{S},j}, \kappa\right)\right] \quad , \tag{7}$$

FSRI_j can be interpreted as the *equity-weighted sum of losses suffered by individual banks in the network*. It is the fraction of overall bank equity that is lost after the initial failure of a single firm propagates through the supply chain network, W, causing firm insolvencies and loan write offs. It can be used to rank the companies operating in the production network by their systemic relevance for financial stability. Note the difference to simply computing the DebtRank of companies in the credit network without their role in the supply chains, as was done in Poledna et al. (2018) and Landaberry et al. (2021).

Fig. 1(c) shows for our example SCN how FSRI_f of firm f is computed as the weighted average of the direct and indirect losses shown in panel (b). The weights are given by the relative equity size of every bank, i.e. $w_3 = w_4 = 0.25 = 1/m$, since we assume that all banks in our example have the same equity. We see that $\mathrm{FSRI}_f = 0.15 = \mathcal{L}(\psi)$, meaning that 15% of the overall bank equity is at risk should firm f fail. Similarly, if we simulate the initial failure of the other firms in the example SCN, we expect system-wide losses of 20%, 15%, 5%, 5%, and 0% for the shock vectors $\psi^{\mathrm{S},c}$, $\psi^{\mathrm{S},d}$, $\psi^{\mathrm{S},b}$, $\psi^{\mathrm{S},e}$, and $\psi^{\mathrm{S},a}$, respectively.

To highlight the effect from supply chain contagion, we distinguish between the *direct* and *indirect* components of FSRI

$$\mathrm{FSRI}_j^{\mathrm{dir}} \equiv \sum_{k=1}^{m} \frac{e_k}{\sum_{l=1}^{m} e_l} \min\left[1, \mathcal{L}_k^{\mathrm{dir}}\left(\psi^{\mathrm{S},j}, \kappa\right)\right] \quad \textit{and}$$
$$\mathrm{FSRI}_j^{\mathrm{indir}} = \mathrm{FSRI}_j - \mathrm{FSRI}_j^{\mathrm{dir}} \quad . \tag{8}$$

For every firm the direct component is proportional to the loans of that firm; the indirect FSRI is equal to the weighted sum of loans of firms that defaulted because of the operational failure of the firm j.

If in Eqs. (4) and (7) χ is substituted by χ_{eq} or by χ_{l}, we get the FSRI triggered by equity or liquidity defaults,

$$\mathrm{FSRI}_{\mathrm{eq},j} \equiv \sum_{k=1}^{m} \frac{e_k}{\sum_{l=1}^{m} e_l} \min\left[1, \mathcal{L}_k\left(\chi_{\mathrm{eq}}(\psi^{\mathrm{S},j}), \kappa\right)\right] \quad \text{and} \tag{9}$$

$$\mathrm{FSRI}_{\mathrm{l},j} \equiv \sum_{k=1}^{m} \frac{e_k}{\sum_{l=1}^{m} e_l} \min\left[1, \mathcal{L}_k\left(\chi_{\mathrm{l}}(\psi^{\mathrm{S},j}), \kappa\right)\right] \quad , \tag{10}$$

respectively.

Estimating individual bank losses from supply chain contagion. To answer the second research question about how much the risk of individual banks is amplified by supply chain contagion, we need to investigate the exposure of banks to losses transmitted by the supply chain network under more general initial shock scenarios. Typical measures used by banks to assess the riskiness of portfolios are *expected loss* (EL), *value at risk* (VaR) and *expected shortfall* (ES) (McNeil et al., 2015). These measures are calculated from a *loss distribution* (McNeil et al., 2015). Here we estimate the loss distribution for each bank with a Monte Carlos simulation that generates initial shock scenarios – representing the simultaneous failure of multiple firms – to the supply chain network. In this way we generate different stress scenarios for the economy and simulate the corresponding losses for each bank.

[7] The super script S indicates that the shock vector, $\psi^{\mathrm{S},j}$, represents the failure of a *single* firm j.

Specifically, we simulate 10,000 initial shocks, $\psi^{\mathrm{M},\ell}$, where multiple firms suffer a full production disruption. The value $\psi_i^{M,\ell} \in \{0,1\}$ indicates the production level of firm i under the initial shock scenario, ℓ. We use the index $\ell \in \{1,2,\dots,10000\}$ to distinguish the 10,000 scenarios. Every shock scenario vector, $\psi^{\mathrm{M},\ell}$, is created by drawing from a n-dimensional multivariate Bernoulli random variable, where the success probabilities correspond to the n-dimensional vector of default probabilities (PDs) of firms, i.e.

$$\psi_i^{\mathrm{M},\ell} \sim \mathrm{Bernoulli}(\mathrm{pd}_i) \quad ,$$

where pd_i is the default probability of firm i. We set the initial production level of firm i to $\psi_i^{\mathrm{M},\ell} = 0$ in case of a success and to 1, otherwise. The PDs are estimated by the Central Bank of Hungary, for details see Appendix C. Note that due to data limitations we draw the defaults of firms independently, i.e. we neglect systematic or correlated events like a large crises affecting specific industry sectors. We denote the set of initial shock vectors by $\Psi^{\mathrm{M}} = \{\psi^{\mathrm{M},1}, \psi^{\mathrm{M},2}, \dots, \psi^{\mathrm{M},\ell}, \dots, \psi^{\mathrm{M},10000}\}$.[8]

As before, contagion leads to direct and indirect losses for banks. For each bank k, we compute the two loss distributions consisting of its *contagion-adjusted* equity losses, $\left(\mathcal{L}_k(\psi^{\mathrm{M},1}), \mathcal{L}_k(\psi^{\mathrm{M},2}), \dots, \mathcal{L}_k(\psi^{\mathrm{M},10000})\right)$ and its *direct* equity losses $\left(\mathcal{L}_k^{\mathrm{dir}}(\psi^{\mathrm{M},1}), \mathcal{L}_k^{\mathrm{dir}}(\psi^{\mathrm{M},2}), \dots, \mathcal{L}_k^{\mathrm{dir}}(\psi^{\mathrm{M},10000})\right)$, corresponding to the 10,000 shock scenarios, Ψ^{M}.

Similarly, $\mathcal{L}(\psi^{\mathrm{M},\ell})$ is used to obtain the loss distribution for the entire banking system. For a given shock vector $\psi^{\mathrm{M},\ell}$, the losses $\mathcal{L}_k(\psi^{\mathrm{M},\ell})$ and $\mathcal{L}_k^{\mathrm{dir}}(\psi^{\mathrm{M},\ell})$ are obtained by performing

- Step 1 for $\psi = \psi^{\mathrm{M},\ell}$ and yielding the production losses of firms, $h(T)$;
- Step 2 yielding the vector of defaulted firms, $\chi(\psi^{\mathrm{M},\ell})$;
- Step 3 yielding the equity losses of banks, $\mathcal{L}_k(\psi^{\mathrm{M},\ell})$, for $\chi(\psi^{\mathrm{M},\ell})$; and $\mathcal{L}_k^{\mathrm{dir}}(\psi^{\mathrm{M},\ell})$ for $\chi^{\mathrm{dir}}(\psi^{\mathrm{M},\ell})$.[9]

The procedure is repeated for every $\ell \in \{1,2,\dots,10000\}$.

For the VaR and ES we choose the 95% threshold. To compare these risk measures for the direct and the contagion-adjusted loss distributions, we use the *risk amplification factor*, ρ_k^X for bank k, where X stands for EL, VaR, or ES. We define it as

$$\rho_k^{\mathrm{EL}} \equiv \frac{\tilde{\mu}_k^{\mathrm{EL}}}{\mu_k^{\mathrm{EL}}} \quad , \quad \rho_k^{\mathrm{VaR}} \equiv \frac{\tilde{\mu}_k^{\mathrm{VaR}}}{\mu_k^{\mathrm{VaR}}} \quad , \quad \rho_k^{\mathrm{ES}} \equiv \frac{\tilde{\mu}_k^{\mathrm{ES}}}{\mu_k^{\mathrm{ES}}} \quad , \tag{11}$$

where μ_k^X is the value calculated from the direct- and $\tilde{\mu}_k^X$ from the contagion-adjusted loss distribution of bank k. We denote the averages over all banks by ρ^{EL}, ρ^{VaR}, and ρ^{ES}. We show how the risk measures and amplification factors vary for different values of κ in Appendix B.

For simulating the effects of the COVID-19 inspired shock we define more general initial shock vector, ψ^C. Here the element, $\psi_i^C \in [0,1]$, represent the fraction of production firm i can maintain in response to the effects of COVID-19. We describe the shock construction in more detail in Section 4.3.

3. Data

We calibrate the model to a unique real-world firm-level data set composed of 4 distinct micro-data sets that are available within the *Central Bank of Hungary*. These are the

1. VAT data based supplier-buyer relationships between practically all firms in Hungary;
2. balance sheets and income statements of firms;
3. loans of banks to firms;
4. CET1 equity of banks.

[8] The super script M indicates that the shock vector, $\psi^{\mathrm{M},\ell}$, represents the simultaneous failure of *multiple* firms.

[9] Finding $\mathcal{L}_k^{\mathrm{dir}}(\psi^{\mathrm{M},\ell})$ is equivalent to a procedure in which step 1, i.e. contagion, is skipped and $h(T)$ is set to be equal to the initial shock, $\psi^{\mathrm{M},\ell}$. Then, it is used in step 2 and χ^{dir} is obtained from Eq. (2).

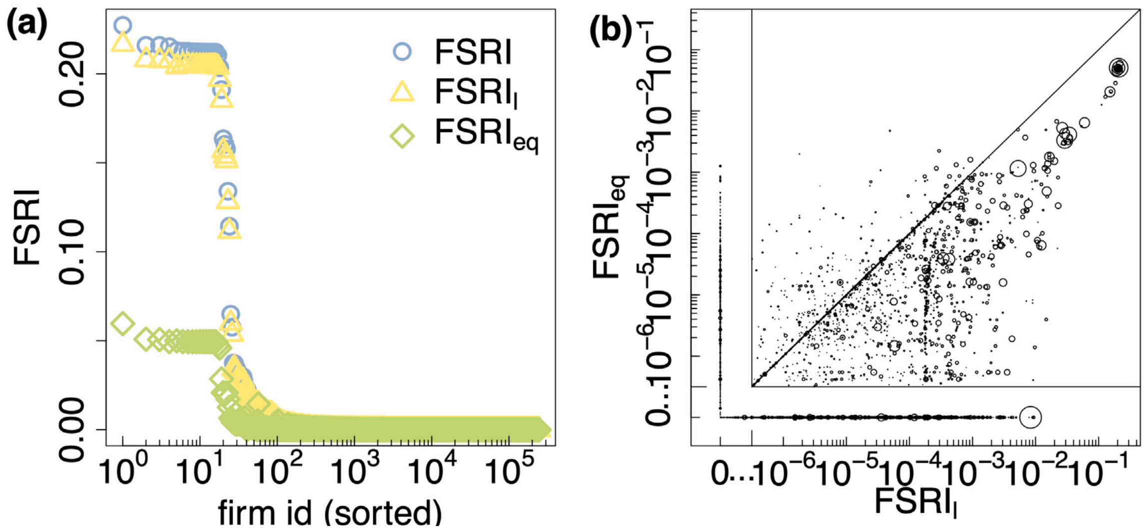

Fig. 2. Financial systemic risk index (FSRI) of firms in the Hungarian SCN. (a) Log-linear scatter plot of FSRI of firms sorted in decreasing order of FSRI (blue circles). The x-axis refers to firms' FSRI-rank from 1 to 243,339, the y-axis represents the FSRI values of those firms. FSRI is shown for two causes of defaults independently: FSRI$_l$ (yellow triangles) - due to negative short-term liquidity and FSRI$_{eq}$ (green diamonds) due to negative equity. FSRI (blue circles) is due to either one of them. All three distributions are sorted in the same way with respect to FSRI. Out of 243,339 firms in the SCN about 100 are systemically risky with 24 (first 24 from left) having very large systemic risk, $10-22\%$ of the overall banks' equity is lost if any of them fails. These companies are the same for all three distributions. (b) Log–log (plus a zero-section) scatter plot of the FSRI$_l$ against the FSRI$_{eq}$, where firms with high values of FSRI$_l$ also have the highest values of FSRI$_{eq}$; see the upper right corner. The size of a bubble is proportional to the out-strength (sales) of a firm. The plot reveals that large systemic risk is not necessarily proportional to the sales of a firm.

The first dataset is used to reconstruct the supply-chain network, W. It is based on the 2019 value added tax (VAT) reports, reflecting any purchase between two firms exceeding 100,000 HUF tax content (approximately 250 Euros) made in Hungary. The resulting network consists of 243,339 anonymized Hungarian companies. The link weights correspond to the monetary value of transactions (price times quantity). We filter the links such that they contain only relationships that are stable. We include links that did occur in at least 2 transactions in two different quarters of the year 2019. 52% of links are stable in this sense and they cover 93% of the monetary value in the network. For statistics on in- and out-degree, and strength distributions, see Appendix D. More information on the dataset, can be found in Borsos and Stancsics (2020), Bacilieri et al. (2022) and Diem et al. (2022a) .

The second dataset consists of the financial statements of the firms in the supply chain network. The variables obtained from the balance sheets and income statements (Table 1) are used to calculate the equity and liquidity buffers of firms, as well as their profits. Out of the 243,339 firms, we do not have financial information on 65,606 firms and exclude another 36,501, because they have either negative equity, liquidity or net income values. Firms with negative equity or liquidity would be a priori in default according to our default definition Appendix B. Firms with negative net income (profits) could experience the counter intuitive effect of gaining equity when having lower operational activity. These firms are still included in the SC contagion and, hence, we can quantify their FSRI, but they cannot default and, hence, cannot cause direct or indirect financial losses. Quantiles of the employed financial variables for excluded and non-excluded firms are shown in Appendix D.

For every firm, we have an empirical estimate of its default probability obtained from a model developed at the Central Bank of Hungary, see Burger et al. (2022) and Appendix C for more details.

The third dataset, the credit registry, is needed to determine the firm-bank loans, B. It consists of the overall exposures of 40,043 firms in the supply chain network, W, to 269 banks in Hungary. The remaining 203,296 companies do not have reported bank loans in Hungary. We remove 242 small-scale banks for which we do not have *Common Equity Tier 1 (CET1)* values (contained in the fourth data set) available, arriving at 27 banks that cover 84% of the loan volume of the original dataset with 269 banks. Hence, B is a matrix of size $n \times m$ with 35,609 non-zero elements (multi-layer links), with $n = 243,339$ and $m = 27$. 32,256 firms have loans at least from one of the 27 banks. Out of the 102,107 excluded firms (negative equity, liquidity, or profits, not data available) 6653 firms have loans that amount to 25% of the total loan volume. Thus, the final number of firms that can cause losses to banks is 25,603. For descriptive statistics of the loan dataset see Appendix D.

Based on this information the model is fully data-driven, meaning that there are no free parameters. The only modelling choices concern the, set-up of the SCN shock propagation (taken from Diem et al. (2022a) and Diem et al. (2024a)), assumptions of how production shocks affect the financial health of a firm, i.e. that revenues and material costs can be reduced proportionally to production losses, the choice that a decrease of 100% in equity or liquidity cause bankruptcy, and the assumption that LGD is equal to 100%, see more in Discussion 5 and Appendix B.

4. Results

4.1. Financial systemic risk of individual firms (FSRI)

Few firms pose high financial systemic risk. We compute the financial systemic risk index (FSRI) (7) for each of the $n = 243,339$ companies contained in the Hungarian supply chain network. The rank-sorted distribution of the FSRI values (blue circles) is shown in Fig. 2(a) in log-linear scale, where the x-axis denotes firm ranks and the y-axis their respective FSRI values. The firm with the highest financial systemic risk is to the very left and the lowest systemic risk firms to the very right. The riskiest firm has an FSRI value of slightly above 22%, meaning that its failure and the ensuing supply chain disruption cascade would lead to loan write offs equivalent to approximately 22% of the overall CET1 capital of the 27 Hungarian banks in the sample (see Fig. E.14(a) for how banks are affected individually). A group of 17 firms form a plateau with very similar FSRI values around 21%. This high-risk plateau is followed by a sharp drop in FSRI values of 7 more firms with values above 10% and 18 firms with values between 10% and 2%. In total, 269 firms have FSRI values higher than 0.1%. To a large extent the relatively high risk of these firms is driven by

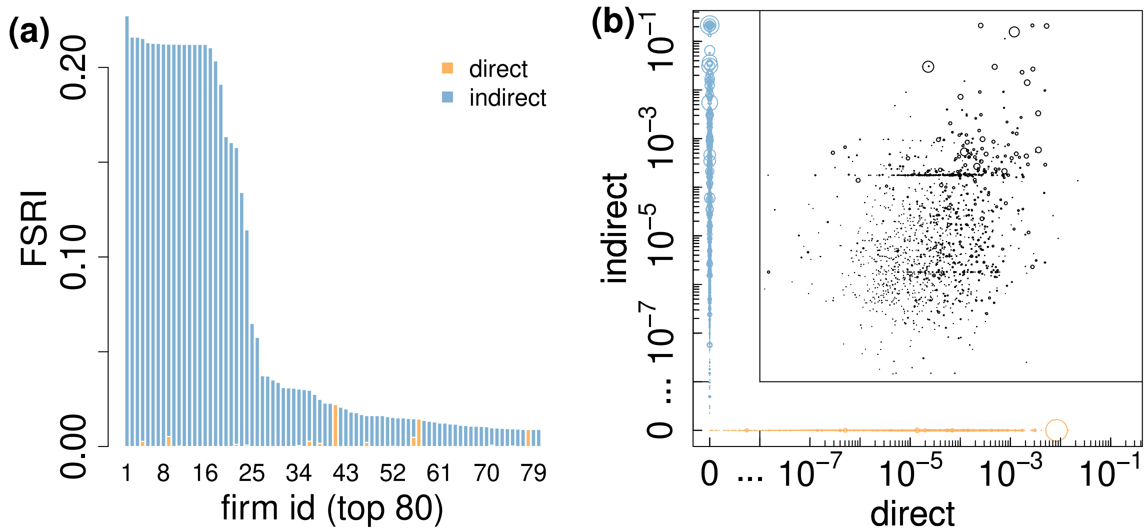

Fig. 3. Direct and indirect components of FSRI. (a) shows the FSRI profile for the first 80 companies, decomposed into the direct and indirect contributions. The direct share $\mathrm{FSRI}_j^{\mathrm{dir}}$ (orange bar) in the FSRI is proportional to company's loans. The indirect share $\mathrm{FSRI}_j^{\mathrm{indir}}$ (blue bar) is proportional to loans of firms that default after j triggers a cascade. It is obvious, that the riskiest firms cause substantial financial losses indirectly through spreading shocks along the SCN. (b) log–log (plus a zero section) scatter plot of direct and indirect FSRI shares of all firms. Circle size is the out-strength. Most large firms typically cause indirect effects. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

the substantial supply chain cascades they trigger (indirect losses). The firms in the high FSRI plateau cause similar supply chain cascades, i.e., they affect mostly the same firms in response to their failure and hence also their financial impacts on the firms in the network is similar, finally causing similar losses to banks' equity levels. This pattern of similar supply chain cascades can be explained by the observed plateau firms, forming a tightly knit network (systemic core of the economy) of highly risky supply relations (Diem et al., 2022a). In Fig. A.9 we study the relation between FSRI and ESRI (economic systemic risk index). ESRI measures the size of the supply chain disruption cascade (in the real economy) in terms of the fraction of the production network's total output that is affected in case of a firm's failure; for details on ESRI and its calculation see Appendix A. As expected, companies from the plateau predominantly belong to the energy, communication, and transportation sectors, a few belong to waste collection and manufacture of basic chemicals. These sectors have been identified as essential to many other industries in the survey conducted by Pichler et al. (2022b).

Financial losses caused by illiquidity dominate losses caused by insolvency. Next we distinguish between the bank capital losses that originate from insolvencies due to a lack of liquidity or a lack of equity. The financial systemic risk index, $\mathrm{FSRI}_{\mathrm{eq}}$ Eq. (9), solely based on firm insolvencies caused by their equity turning negative after supply-chain shock propagation is shown by green diamonds. Similarly, the financial systemic risk index, $\mathrm{FSRI}_{\mathrm{l}}$ Eq. (10), solely based on firm insolvencies caused by their liquidity turning negative after supply-chain shock propagation is shown by yellow triangles. As expected, $\mathrm{FSRI}_{\mathrm{eq}}$ and $\mathrm{FSRI}_{\mathrm{l}}$ are consistently smaller than FSRI, since the insolvency criterion for FSRI encompasses both, equity and liquidity based insolvencies. As expected, the high-risk plateaus of $\mathrm{FSRI}_{\mathrm{eq}}$ (5%) is lower than the liquidity default induced plateau, $\mathrm{FSRI}_{\mathrm{l}}$, (20%). This indicates that for the events (firm failures) causing the largest amounts of supply-chain contagion, the majority of defaults and bank equity losses are caused by firms turning illiquid. Most firms with equity problems would have liquidity problems already. In the $\mathrm{FSRI}_{\mathrm{eq}}$ profile distribution there are 18 firms with risk over 4% (with maximum at 6%), additional 7 firms in the interval $1-4\%$, and in total 73 firms with risk higher than 0.1%. Similarly, in the $\mathrm{FSRI}_{\mathrm{l}}$ profile distribution there are 18 firms with risk over 19% (with maximum at 21%), additional 6 firms in the interval $10-19\%$, and a total of 252 firms exceeding the risk of 0.1%.

To investigate the relation between liquidity- and equity based insolvencies more closely, we provide a scatter plot in log–log scale in Fig. 2(b). Since, many firms only cause either liquidity or equity based insolvencies, we plot two additional segments that would not be visible on the log–log scale. The first one shows the $\mathrm{FSRI}_{\mathrm{eq}}$ values if $\mathrm{FSRI}_{\mathrm{l}} = 0$ and the second segment shows the $\mathrm{FSRI}_{\mathrm{l}}$ values if $\mathrm{FSRI}_{\mathrm{eq}} = 0$. Bubble size is proportional to the out-strength (sales to other firms proportional to revenues) of the respective firm (see Appendix A). As can be seen from the top right corner, plateau firms of $\mathrm{FSRI}_{\mathrm{eq}}$ coincide with plateau firms of $\mathrm{FSRI}_{\mathrm{l}}$. Interestingly, large firms (size of bubble) are not necessarily systemically important ones, and, on the other hand, some small firms have large systemic importance. On correlation of FSRI with output and loan sizes of firms see Appendix E. For small values $\mathrm{FSRI}_{\mathrm{l}}$ and $\mathrm{FSRI}_{\mathrm{eq}}$ seem uncorrelated. There are 21,282 firms with non-zero FSRI values, 19,227 for $\mathrm{FSRI}_{\mathrm{l}}$ and 10,216 for $\mathrm{FSRI}_{\mathrm{eq}}$. The low number of positive FSRI values can be explained by two factors. First, even though economic systemic risk (ESRI) – measuring the fraction of the production that is lost in the supply chain network after the failure of a firm – values decay as power law (see Diem et al. (2022b) Fig. S4), only a few firms cause large supply-chain cascades that can affect the financial conditions of other firms severely enough to cause a default. Second, in the data set 13% of firms have loans, i.e., only 32,256 firms out of 243,339 have loans in one of the 27 banks. This number is further reduced to 25,603 firms that have non-negative equity, liquidity and profits before the shock (see Appendix B). Hence, indirect losses to banks are only caused when supply chain contagion leads to default of these firms.

High systemic risk firms cause predominantly indirect financial losses. We next investigate the relation between losses caused directly by firms defaulting on their own loans and indirect loan defaults caused by secondary supply chain disruptions. Fig. 3(a), disaggregates the FSRI values into direct and indirect losses for those 80 firms that cause the largest bank equity losses. The direct share $\mathrm{FSRI}_j^{\mathrm{dir}}$ (orange bar) of firm j (see Eq. (8)) is simply the size of its defaulted loans divided by overall banks' equities. Naturally most loans are very small with respect to total bank equity. The indirect share $\mathrm{FSRI}_j^{\mathrm{indir}}$ (blue bar) defined in Eq. (8) is proportional to loans that default after j triggered a supply-chain cascade. The overwhelmingly blue colour in the bar-plot indicates that the highest losses to overall bank equity are caused by supply chain

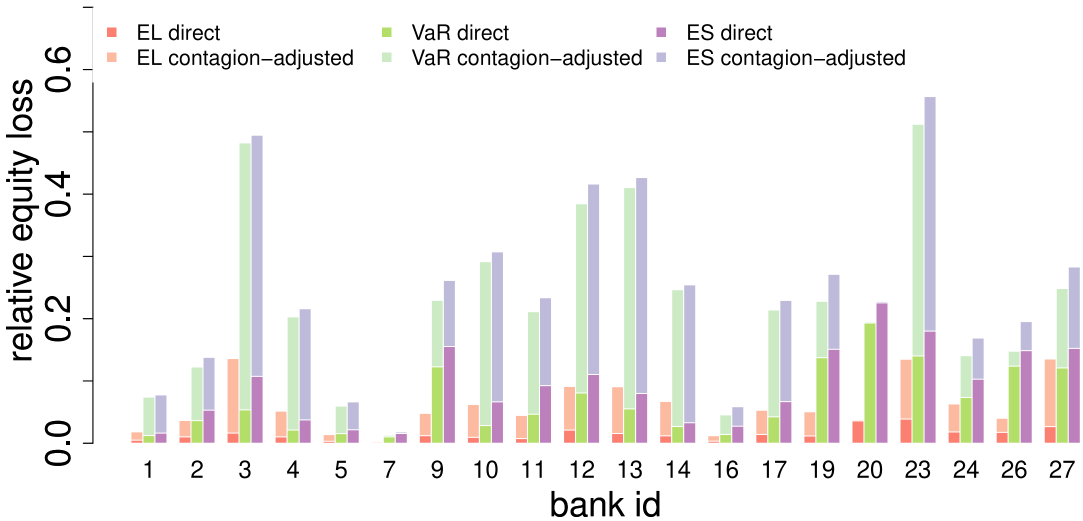

Fig. 4. Expected losses, value at risk, and expected shortfall of banks including only direct and contagion-adjusted equity losses of banks. Indices of banks (x-axis) are the same as in Fig. E.14. Bank losses on y-axis are given with respect to individual equity of each bank. We omit banks 6, 8, 15, 18, 21, 22, and 25 that have less than 30 clients. Loss distributions are generated from relative equity losses, $\mathcal{L}_k(\Psi^M)$ from Eq. (4), for 10,000 shock scenarios. The risk measures EL, VaR, and ES are shown in red, green and purple, respectively; direct losses in dark shades, contagion-adjusted losses in light shades. The bar-plot is not stacked, dark bars are plotted in front of light bars. Taking into account the exposure to the SCN massively amplifies risks for all banks. The average risk amplification factors are $\rho^{\mathrm{EL}} = 5.2$, $\rho^{\mathrm{VaR}} = 6.7$, and $\rho^{\mathrm{ES}} = 4.4$. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

shock propagation and not by the size of the firms' own loans, see also Fig. E.13. A direct loss can be caused only by firms that have loans. If a firm without a loan initially fails and defaults it will not cause a direct loss. For instance, out of the 80 firms in Fig. 3(a) 35 have loans and 41 default. The two sets intersect for 19 firms, whereas the remaining 16 firms with loans do not default because of their strong equity and liquidity buffers, i.e., they cause only indirect losses. In Fig. 3(b) we show the relation of direct versus indirect losses with a log–log (plus a zero section) scatter plot. In more detail, 16,183 companies have only positive direct effects, i.e $\mathrm{FSRI}_j = \mathrm{FSRI}_j^{\mathrm{dir}}$. 6553 firms (blue circles) have only positive indirect effects, i.e. $\mathrm{FSRI}_j = \mathrm{FSRI}_j^{\mathrm{indir}}$. The remaining 1454 firms (black circles) have non-zero direct as well as indirect effects, i.e. $\mathrm{FSRI}_j = \mathrm{FSRI}_j^{\mathrm{dir}} + \mathrm{FSRI}_j^{\mathrm{indir}}$. Most of the biggest companies, with respect to out strength (bubble size) cause zero direct losses (blue circles). Nevertheless, in Fig. 3(a) we see some firms with big loans causing small indirect effect. In particular, firms 41, or 78 (showing two highest orange bars) have loans size around 2% and 1% of total banks' equities, and belong to the top 4% based on their sale sizes.

4.2. Exposure to supply chain contagion of individual banks

Supply chain contagion amplifies risk for banks. We next focus on how the credit risk exposure of individual banks is amplified in the presence of supply chain contagion (for an example of losses to individual banks from single firm's failure see Appendix E). We measure the amplification by calculating the EL, VaR, and ES for the simulated losses of each banks' commercial loan portfolio, once without supply chain contagion (direct losses only) and with supply chain contagion. We calculate the direct losses, $\mathcal{L}_k^{\mathrm{dir}}(\psi^{\mathrm{M},\ell})$, and the supply chain contagion adjusted ones, $\mathcal{L}_k(\psi^{\mathrm{M},\ell})$ with a Monte Carlo simulation for 10,000 initial shock scenarios, $\ell \in \{1,\dots,10\,000\}$, as described in Section 2, step 3. EL is the mean of the losses, $\mathrm{VaR}_{0.95}$ is the 95% quantile of losses, and the $\mathrm{ES}_{0.95}$ is the average over the 500 largest losses. Fig. 4 shows the EL (red), VaR (green), and the ES (purple) of every bank based on the direct (dark shade) and the contagion-adjusted (light shade) loss distributions. Indices of banks are the same as in Fig. E.14. We skip banks 6, 8, 15, 18, 21, 22, and 25 that have less than 30 clients. The y-axis shows the value of the respective risk measures. Note that the bars are not stacked (dark bars are always plotted in front of light bars) meaning that the height of the light shaded bars is the overall value of the risk measure with supply chain contagion. The height of the light bars minus the height of the dark bars gives the additional risk from contagion (marginal systemic risk). The plot shows that EL, VaR, and ES are substantially lower when the supply chain is not taken into account. The average risk amplification factor across the 20 banks are found to be $\rho^{\mathrm{EL}} = 5.2$, $\rho^{\mathrm{VaR}} = 6.7$, and $\rho^{\mathrm{ES}} = 4.4$. Note that the overall levels of credit risk from commerical loans strongly differ across banks — again potential reasons being differences in business models, etc. as outlined before. Amplification factors also differ across banks.

Tail losses are driven by defaults of high systemic risk firms. Finally, in Fig. 5 we present the system-wide histogram of direct losses, $\mathcal{L}^{\mathrm{dir}}(\psi^{\mathrm{M},\ell})$ (orange) and the contagion-adjusted ones, $\mathcal{L}(\psi^{\mathrm{M},\ell})$ (brown), summed over all 27 banks; see Eq. (5). The x-axis denotes the fraction of equity that is lost due to an initial shock scenario, $\psi^{\mathrm{M},\ell}$, the y-axis shows the frequency of the respective loss. One clearly sees that the two distributions differ substantially. The direct loss distribution is concentrated at low values with maximum around 4%. The contagion-adjusted loss distribution is bimodal with most weight centred around 1.7%. It slowly decays with a fat tail and has a second concentration point around 23% of system wide equity losses. In between there are a few scenarios yielding losses of around 7% and 13%. This pattern is driven by the shape of the FSRI distribution; see Fig. 2. The most risky firms cause system-wide losses of around 21% of equity. If one of these firms fails in an initial shock scenario, $\psi^{\mathrm{M},\ell}$, they trigger a supply chain cascade leading to losses of around 21%. Similarly, there are a few high systemic risk firms causing losses of around 13% and 7% of equity, explaining the respective parts in the loss distributions. The risk respective amplification factors for EL, VaR, and ES (of contagion-adjusted and direct loss distributions) are 6.4, 16.1, and 12.2, respectively.

4.3. Financial stability effects of a COVID-19 shock and cost efficient policy interventions

Now we demonstrate the practical usefulness of our method in assessing risks to financial stability posed by large crisis. Our demonstration leverages the initial economic shock caused by the COVID-19 pandemic in 2020 Q1. COVID-19 is an ideal showcase as supply chain contagion played a major role, and it initially triggered large concerns

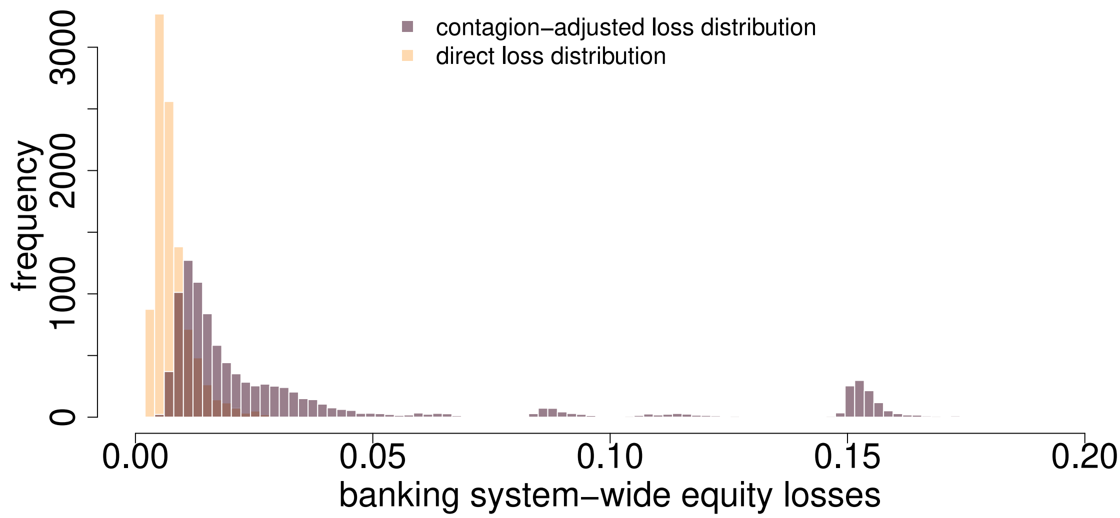

Fig. 5. Direct (orange) and contagion-adjusted (brown) loss distributions across the entire banking layer of 27 banks. The x-axis shows losses to the financial system, relative to the overall equity of all banks, the y-axis is the occurrence frequency of losses. The peak around 0.22 corresponds to shock scenarios in which firms with highest FSRI default. Risk amplification factors for the EL, VaR, and ES are 6.4, 16.1, and 12.2, respectively. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

about the effects on financial stability. For example, the empirical evidence on Italian firms described in Schivardi and Romano (2020) suggests that the effects of the pandemic on the financial health of firms were substantial (around 180,000 firms with 3.1 million workers became illiquid already in April 2020). This unprecedented event required a quick response from policy makers to provide additional liquidity to firms to avoid an unnecessary large number of defaults. The here proposed framework can be used to quickly receive a detailed picture of the financial losses of individual firms due to supply chain contagion and how these losses affect firms liquidity and equity buffers, loan write offs and ultimately financial stability. The granularity of the simulation framework allows us to test the effects of policy intervention to reduce the threats to financial stability.

4.3.1. Financial system losses of an empirically calibrated and synthetic COVID-19 shocks

In our model the initial shock from COVID-19 is modelled as the shock to firms' production levels in the first quarter of 2020, and is calibrated from monthly firm-level employment data. For the design of the initial production shock, $\psi^{\mathrm{C}} \in [0,1]^n$, we follow Diem et al. (2024a). The data (collected by the Hungarian tax authority) contains monthly information about the number of employees for 160,000 out of all 243,339 firms. We assume that the shock to firms production levels is proportional to the relative reduction in firms' intensity of the input labour between January to May 2020. Thus, we assume that if firm, i, lost 30% of employees, its production level initially (before SCN-contagion) drops to $h_i(t_1) = \psi_i^{\mathrm{C}} = 0.7$. In general, we define the production level of firm i, $h_i(t_1) = \psi_i^{\mathrm{C}}$, at t_1 as

$$\psi_i^{\mathrm{C}} = \max\left[1 - \frac{empl_i(\mathrm{May})}{empl_i(\mathrm{Jan})}, 0\right] \quad , \tag{12}$$

where $empl_i$(month) is the number of employees of firm i in a given month.[10] For firms j with no employment data we impute the shock, ψ_j^{C}, by drawing a value across all shocks, ψ_l^{C}, of firms l that have the same NACE4 category as j and for which employment data was available. We repeat this imputation procedure 1000 times and take the shock vector that corresponds to the median of the SC-contagion induced losses of the 1,000 shocks. For 64,474 firms (with imputation) in the given period between January and May 2020, the number of employees dropped, i.e., received a shock. We define the production shock (with imputation) given by this data as the *empirical COVID-19 shock*, $\psi^{\mathrm{C,emp}}$. As estimating how a crisis is initially affecting the production levels of individual firms correctly is hard and supply chain contagion is sensitive to which exact firms are suffering an initial shock, we can sample multiple initial shocks and receive a distribution of system wide financial losses. To do so we generate 1000 synthetic COVID-19 shock scenarios, $\psi^{\mathrm{C},\ell}$ where $\ell \in \{1,2,\dots,1000\}$ initial shocks that have the same shock size at the level of industry sectors (NACE 2 digit), but affect firms within these sectors differently.[11] Hence, the synthetic shocks allow us to reflect the uncertainty of how the exact initial shock materialized, while keeping the overall economic size of the shock constant. As for production losses (Diem et al., 2024a), simulating the losses of the synthetic shocks demonstrates the additional value of our firm-level shock propagation framework, as for models using aggregate input–output data these 1000 synthetic shocks would all yield the exact same SCN-contagion and hence financial system losses. In this way sector level models can underestimate threats to financial stability posed by SCN-contagion, because they cannot take into account if systemically important firms received a shock or not.

Financial losses are substantially amplified due to SCN-contagion. Fig. 6 shows the distribution of the system wide financial losses for the empirically calibrated COVID-19 shock, ψ^{C}, (vertical lines) and the 1000 synthetic shocks, $\psi^{\mathrm{C},\ell}$ where $\ell \in \{1,2,\dots,1000\}$, (histograms). For the empirical COVID-19 shock, the losses from firms turning insolvent (dashed line) are $\mathcal{L}\left(\chi_{\mathrm{eq}}(\psi^{\mathrm{C,emp}})\right) = 0.011$, losses from firms turning illiquid (dotted line) are $\mathcal{L}\left(\chi_{\mathrm{l}}(\psi^{\mathrm{C,emp}})\right) = 0.056$, and losses from either insolvency or liquidity (solid line) are $\mathcal{L}\left(\chi(\psi^{\mathrm{C,emp}})\right) = 0.062$, i.e., 6.2% of overall bank equity is lost in response to the COVID-19 shock. The initial shock to firms production is responsible for 3% of system wide financial losses and indirect losses due to supply chain contagion for 3.2%. Hence, for the COVID-19 shock the financial system losses are amplified by more than 100% (a factor

[10] Note that here we focus only on negative shocks, as the overwhelming majority of sectors was negatively affected, and demand increased for only few sectors (del Rio-Chanona et al., 2020).

[11] For the detailed description of the algorithm for generating synthetic shocks, see Diem et al. (2024a).

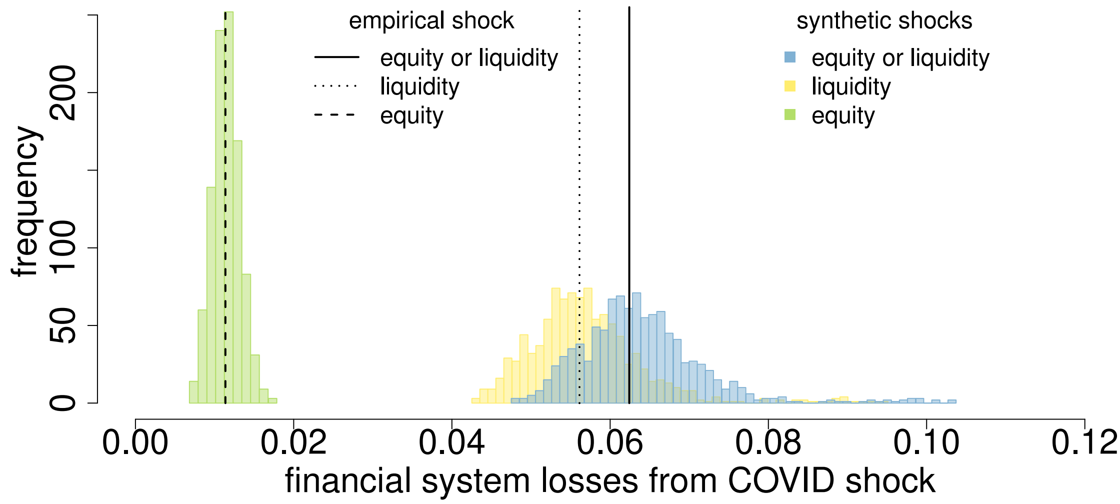

Fig. 6. Banking system equity losses from COVID-19 shock. Banking (financial) system equity losses defined by (5) are on x-axis, y-axis denotes a frequency of occurrences. Histograms are filled with $\mathcal{L}\left(\chi_{\mathrm{eq}}(\psi^{\mathrm{C},\ell})\right)$ (green), $\mathcal{L}\left(\chi_{\mathrm{l}}(\psi^{\mathrm{C},\ell})\right)$ (yellow) and $\mathcal{L}\left(\chi(\psi^{\mathrm{C},\ell})\right)$ (blue) where $\ell \in \{1,2,\ldots,1000\}$ represents 1000 synthetic COVID-19 shocks. Vertical lines indicate losses from empirical COVID-19 shock: $\mathcal{L}\left(\chi_{\mathrm{eq}}(\psi^{\mathrm{C,emp}})\right) = 0.011$, $\mathcal{L}\left(\chi_{\mathrm{l}}(\psi^{\mathrm{C,emp}})\right) = 0.056$ and $\mathcal{L}\left(\chi(\psi^{\mathrm{C,emp}})\right) = 0.062$. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

of 2.1) due to SCN-contagion. In the empirical COVID-19 shock the initial production losses of firms sum up to 4.8% of total out strength ($\sum \psi_i^{\mathrm{C,emp}} s_i^{\mathrm{out}} = 0.048$) and supply chain contagion causes an additional loss of production of 6.6% of total out-strength (indirect loss).

For the 1000 synthetic COVID-19 shocks, the losses from firms turning insolvent (green histogram) $\mathcal{L}\left(\chi_{\mathrm{eq}}(\psi^{\mathrm{C},\ell})\right)$, range from 0.007 to 0.018, losses from firms turning illiquid (yellow histogram) $\mathcal{L}\left(\chi_{\mathrm{l}}(\psi^{\mathrm{C,emp}})\right)$, range from 0.043 to 0.095, and losses from either insolvency or liquidity (blue histogram) $\mathcal{L}\left(\chi(\psi^{\mathrm{C,emp}})\right)$, range from 0.047 to 0.104.[12] Average supply chain amplification factors (total system losses divided by direct system losses) are 1.61 (green), 1.72 (yellow) and 1.73 (blue). The average losses of the synthethic shocks equal 0.011, 0.057 and 0.064, for insolvency, illiquidity, and either insolvency or illiquidity, respectively. Due to the design of the synthetic shocks the averages are close to the losses caused by the empirical shock. The distribution of system wide financial losses from insolvency are substantially less spread out than the one for illiquidity, i.e., which exact firms are affected by the initial shock matters more for firms liquidity than solvency (equity levels). The losses taking both types of default into account (blue histogram) are very heavy tailed, i.e., in some initial shock scenarios large losses are possible (even though the aggregate size across initial shocks is the same). Overall it is clearly visible that the defaults from illiquidity dominate over losses caused by firms insolvency (negative equity buffers).

Note that the COVID-19 shock induced losses are substantially smaller than the losses caused by the failure of the high FSRI plateau firms, as none of the plateau firms suffers a 100% shock. In the empirical COVID-19 shock, $\psi^{\mathrm{C,emp}}$, two of the plateau firms are hit by 33% and 24% initial shock, respectively. There are another 8 firms from the plateau that suffer a less then 2% shock. Note that for the PD-based monte-carlo simulations the mean amplification factor across 10,000 simulations (Fig. 5) is 7.0, which is substantially higher than for the COVID-19 shock. However, the median amplification factor across the 10,000 initial shocks is 2.3, i.e. very similar to one of the COVID-19 shock. The mean amplification factor is much higher due to the large tail of indirect losses caused by the failures of high systemic risk firms. In Appendix G, Fig. G.20, we additionally show how the synthetic COVID-19 shocks affect risks of separate banks.

Financial losses stemming from industrial sectors are not proportional to the production losses they suffer. We now show for the empirical COVID-19 shock, from which industry sectors the direct and indirect system wide financial losses originate (Fig. 7(a)), and compare them to the direct and indirect production losses firms within the sectors suffer (Fig. 7(b)). From Fig. 7(a) it is clearly visible that almost 50% of overall financial losses stem from loan write offs of firms in sector C, manufacturing, and these are pre-dominantly caused (indirectly) by supply chain contagion and not the initial shocks. Note that in Fig. 7(b) we can observe that the manufacturing sector only suffered around a fifth of the overall production losses. In comparison, sector G, whole sale and retail trade, suffers almost the same amount of overall production losses as the manufacturing sector, but is responsible for financial system losses of only 0.075. Hence, financial system losses of sectors are not simply proportional to production losses they suffer, but a non-trivial function of the amount of loans across the sector, and the distribution of financial buffers (equity and liquidity) of firms in the sector. Details on distributions of direct and indirect production and financial losses of the synthetic COVID-19 shocks across the NACE1 sectors are discussed in Appendix G.

4.3.2. Testing (optimal) policy interventions to reduce threats to financial stability

After assessing how large the (financial) losses due to an emerging crisis are, regulators and policy makers must decide if a policy intervention is necessary to maintain financial stability, and if so, how the policy intervention should be designed (i.e., which firms receive, which type of support, and how much of it). We test two types of (financial) support policies – supplying firms with additional liquidity or providing equity subsidies – for dampening the effects of the COVID-19 shock on financial stability.[13] Both types of measures have been very actively deployed by governments. Often such stimulus packages were not very targeted (some firms needed more, some needed less than they got), e.g., there is ample evidence of firms that running high profits while receiving large government subsidies, and the number of firm defaults

[12] Amplification factors defined as system losses against direct losses of min (max) values of green, yellow and blue histograms are 1.6 (1.4), 1.4 (1.9) and 1.4 (2) respectively.

[13] Note that technically governments could prop up demand of firms by buying goods, but this could take strange turns e.g., buying directly from restaurants. Providing additional supply of scarce products is even harder.

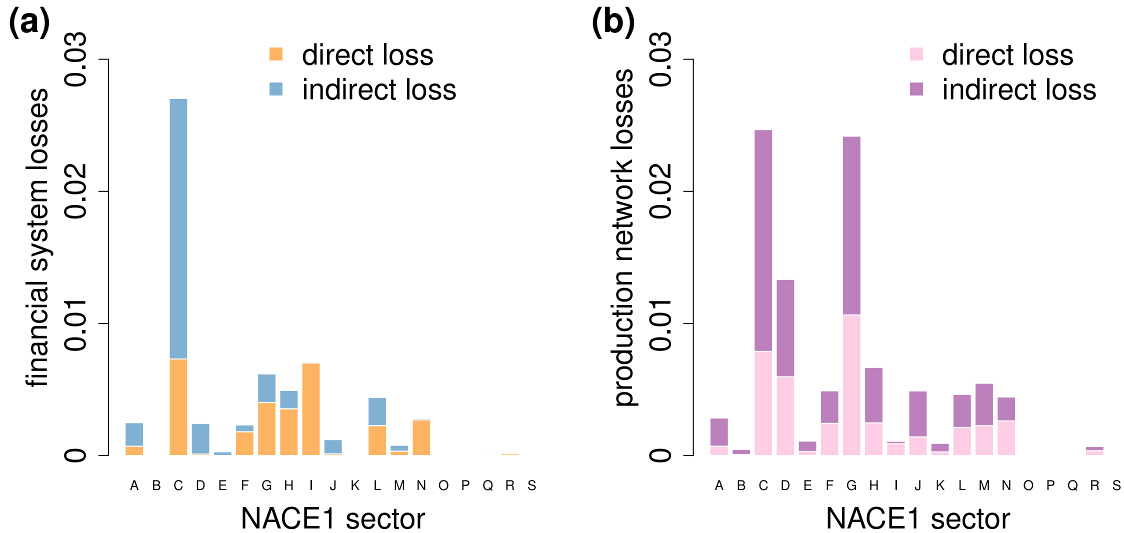

Fig. 7. Direct and indirect financial and production systemic losses initiated by empirical COVID-19 shocks disaggregated to NACE1 sectors. In both plots, letters on the x-axis refer to the top level Nomenclature of Economic Activities (NACE 1). (a) y-axis shows banking system equity losses, orange colour denotes direct losses (caused by initially defaulting firms) and blue colour indicates indirect losses (additional losses from defaults caused by the SCN contagion). (b) y-axis shows systemic production losses equal to output losses of firms given by Eq. (A.5). Pink colour denotes direct losses (initial production shock) and purple colour indicates indirect losses (additional production losses after the SCN contagion). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

dropped drastically in 2020 and 2021 even though there was a large recession (usually during a recession the number of defaults should increase) hints at over-subsidizing firms. Further, over-sized subsidies can have undesirable side effects like creating inflationary pressure. Here we show for a given crisis scenario and a given government budget for liquidity support and equity subsidies, how support measures can be provided in a targeted way such that financial losses are minimized for a given stimulus budget.

Generally there are equity and liquidity support based measures and firms can be either illiquid and solvent, illiquid and insolvent, insolvent and liquid, solvent and liquid. Supporting illiquid yet solvent firms makes sense as they are actually financially sound and only face a short term financing problem, whereas supporting insolvent firms amounts to give money to firms that might have structural problems and, hence, might simply fail just later when the subsidies are used up. Supporting illiquid yet solvent firms is common during crisis e.g., Capponi and Chen (2015) show how a lender of last resort (Bagehot, 1873) can optimally support illiquid banks during crisis. Consequently, we analyse two types of policies: Policy 1 — liquidity support of solvent firms — and Policy 2 — liquidity support plus equity subsidies for insolvent firms. Note that Policy 1 can either be provided through loans, guarantees from the government to secure loans, or as subsidy, while, for Policy 2 actual capital must be provided e.g., in return for company shares or as a subsidy. To do so, we define the liquidity gap of an illiquid (yet solvent) firm i, γ_i^{liq}, as the necessary amount of cash it needs to be able to repay its short term liabilities in the stress scenario, ψ. It is calculated as the difference between profit loss and current liquidity, $\gamma_i^{\text{liq}} = \Delta p_i(\psi) - (a_i - s_i)$. Similarly, the equity gap of an insolvent (potentially liquid) firm i, γ_i^{eq}, is defined as the necessary amount of equity it needs to turn its equity level positive during the stress scenario, ψ, i.e., make it economically viable (at least in the short run). It is calculated as the difference between the profit reduction and the current equity level, $\gamma_i^{\text{eq}} = \Delta p_i(\psi) - z_i$. We define the financial gap of a firm i relative to total bank equity as

$$\gamma_i(\psi) \equiv \frac{1}{\sum_{k=1}^{m} e_k} \max\left[\Delta p_i(\psi) - (a_i - s_i), \Delta p_i(\psi) - z_i, 0\right] \quad , \tag{13}$$

i.e., the smallest amount of money such that the default of firm i can be avoided. The normalization with total bank equity is used below to compare the costs of saving the firm with the losses its default cause to overall bank equity. Note that if a firm is liquid and solvent its financing gap is 0, if it is solvent yet illiquid, γ_i corresponds to the liquidity gap and analogously for liquid yet insolvent to the equity gap, and for illiquid and insolvent it is $\max\left[\gamma_i^{\text{liq}}, \gamma_i^{\text{eq}}\right]$. The total amount of financial resources (normalized with respect to the total equity of banks) that is needed (e.g., by the government) to avoid the defaults of a set, S, of firms after a shock ψ is calculated as

$$\mathcal{G}(S, \psi) = \sum_{i \in S} \gamma_i(\psi) \quad . \tag{14}$$

Loss reduction per unit of financial support varies vastly across firms. In Fig. 8(a–b) we compare the financial gaps of individual firms with the overall bank equity losses due to loan write offs in case the respective firm defaults (both axis are denoted as a fraction of total equity of banks). The financial gaps, γ_i, of firms are on the x-axis and the losses they cause on the y-axis. Note the double log-scale. In total, for the empirical COVID-19 shock scenario, ψ^C, our model yields 2101 firm defaults. Fig. 8(a) shows the comparison for the 1397 illiquid but solvent firms (blue circles). For firms above the diagonal the losses of their default are higher than the amount of liquidity provision required to prevent their default (illiquidity), firms below it need more liquidity support than the losses that would be reduced when they are saved. The further a firm is above the diagonal the more attractive it is to provide it with liquidity. We see that most firms lie above the diagonal (1114 in (a) and 384 in (b)) and the rest (283 in (a) and 320 in (b)) lies below it. There are several orders of magnitude difference in the efficiency (reduced losses per unit of financial support) of supporting a firm, i.e., depending on which firms receive liquidity support outcomes in terms of savings will be tremendously different for a given liquidity support budget. Fig. 8(b) shows the comparison of costs of saving 704 insolvent firms (green triangles) versus the cost implied by their defaults. Note that it is only efficient to support firms above the diagonal (at least when the support is paid out as subsidy, else the decision criterion can be different). Based on the efficiency criterion we can rank firms such that the financial system losses are minimized for a given support budget.

80% **of system wide losses can be reduced with a small liquidity support budget.** Given the firm-level information on costs and gains of saving individual solvent and insolvent firms for a given crisis scenario, we can define a ranking in which order they should be provided with financial support. In Fig. 8(c) we show the cumulative costs of financial

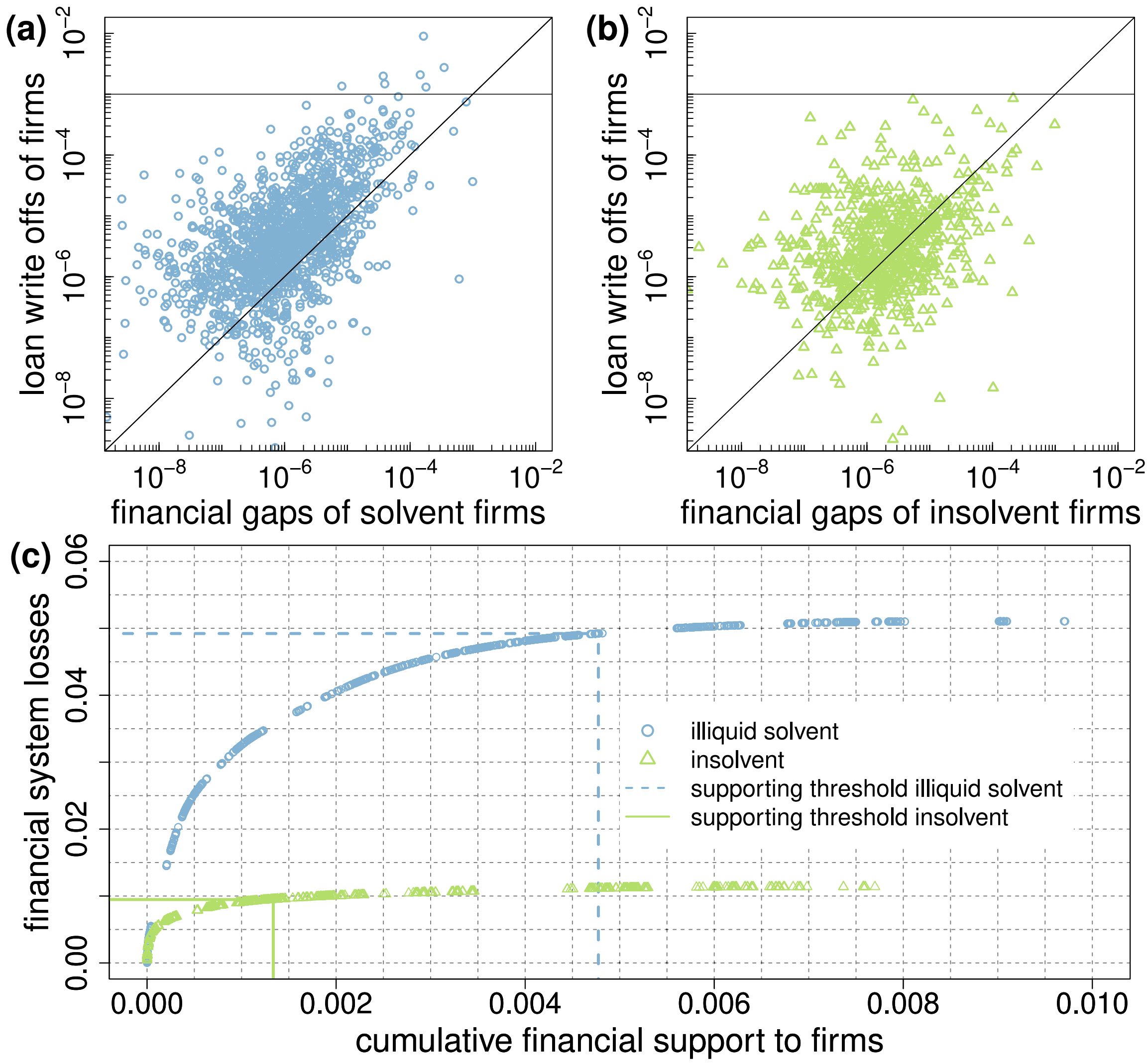

Fig. 8. Financing support to firms in empirical COVID-19 shock scenario, $\psi^{C,emp}$. Panel (a) shows (log–log scaled) results for illiquid solvent firms (blue circles): $\gamma_i(\psi^{C,emp})$ is on the x-axis and loans of those firms as a fraction of total equity of banks on the y-axis. Firms that are above diagonal have bigger loans than financial gaps. The horizontal line indicates in 0.001 y-coordinate. There are 7 firms above that line that together hold loans equal to 2% of equity of banks. Panel (b) shows (log–log scaled) results for illiquid solvent firms (green triangles): $\gamma_i(\psi^{C,emp})$ is on the x-axis and loans of those firms as a fraction of total equity of banks on the y-axis. Firms that are above diagonal have bigger loans than financial gaps. The horizontal line indicates in 0.001 y-coordinate. None of the insolvent firms in empirical scenario has loans higher than 0.1% of equity of banks. Panel (c) shows cumulative financing support against cumulative financial losses simulating two contra-factual situations: with and without assistance. x-axis shows cumulative financing support and the y-axis indicates cumulative financial losses (loan write offs). Blue circles and green triangles correspond to the summation over coordinates of firms in panels (a) and (b) respectively. The right-most points indicate assistance to all firms in the two groups. Lines show the so-called supporting thresholds until which summation is done over firms with gaps smaller than their loans (firms above the diagonals in panels (a) and (b)). After the threshold, summation continues over firms under the diagonals. Points behind the respective thresholds represent sub-optimal policies. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

support on the x-axis and the cumulative avoided financial system losses on the y-axis, for illiquid yet solvent firms (blue circles) and insolvent firms (green triangles) separately. Firms with the highest amount of losses avoided per unit of support are saved first, i.e., the most left circle (triangle) corresponds to saving the firm with the highest saving to cost ratio, the most top right circle (triangle) shows the cost of saving all solvent (insolvent) firms. The dashed blue (solid green) vertical line indicates the cost of saving all firms above the diagonal in Fig. 8(a,b) and the horizontal lines the corresponding losses avoided from defaults. We see that saving all illiquid solvent firms above the diagonal reduces financial system losses of about 4.92% of total bank equity at the cost of providing liquidity support worth only 0.47% of total bank equity. Hence, most financial system losses can be avoided by investing a tiny fraction of bank equity in liquidity support of *solvent* firms. Saving all insolvent firms above the diagonal would reduce the financial system losses by 0.95% of total bank equity at the cost of providing financial subsidies worth 0.13% of total bank equity. Hence, for saving solvent firms the ratio of avoided loss by costs is 10.31 and for saving insolvent firms 7.1, i.e., saving solvent firms is substantially more cost efficient. Note that this comparison does not take into account that the liquidity support can be given as loans (i.e., will be paid back) and that the equity support required for saving insolvent firms will not be simply paid back. The numbers are summarized in Table 2. Note that supporting firms below the diagonal in Fig. 8(a) and (b) is not efficient.

Note that here all figures are based on assuming a LGD of 100%. Naturally, changing the assumed LGD changes the size of the system losses, as shown in Appendix B, the financial system losses change proportionally to LGD. Hence, for lower LGD rates the effectiveness ratios will be lower. It is easy to infer for which values of κ system losses will be smaller than the assistance. For instance, in the optimal strategy for illiquid solvent firms $1/10.31 = 0.097$, thus for $\kappa < 0.097$ the support would be higher than losses. In case of the sub-optimal strategy for insolvent firms the support would be inefficient for $\kappa < 0.67$.

Table 2
Comparison of financing support strategies in empirical COVID-19 shock example. In optimal strategy financing assistance (Fig. 8(c) vertical lines) is provided only to firms that have smaller financial gaps than loans (firms above the diagonals in Fig. 8). In the sub-optimal strategy all the firms get the support to cover their financial gaps (right-most points in Fig. 8(c)). Support prevents system losses and effectiveness is given by the ratio of these two. If the ratio is smaller than 1, support is inefficient. All the strategies presented are efficient, nevertheless they differ in cost-effectiveness. The latter is highest in case of optimal support strategy for illiquid solvent firms.

	Policy 1 (illiquid & solvent)		Policy 2 (insolvent)	
	Optimal selection	All firms	Optimal selection	All firms
Financing support ($\mathcal{G}$)	0.0047	0.0097	0.0013	0.0077
System losses ($\mathcal{L}$)	0.0492	0.0511	0.0095	0.0114
Effectiveness ($\mathcal{L}/\mathcal{G}$)	10.31	5.26	7.10	1.48

5. Discussion

A thorough assessment of credit risk is central to a sustainable banking sector. Classical credit rating models hitherto do not include information about supply chain networks (SCNs) on the firm level, their structures, and default dynamics in a quantitative way. Here we introduced a fully data-driven model (containing a minimum number of free parameters) that allows us to quantitatively estimate how SCN contagion translates to capital losses in the banking sector and to what extent it affects financial stability.

The model is a 1:1 agent-based representation of (almost) every single firm and bank in the national economy of Hungary in 2019, containing more than 240,000 firms, 27 banks, including their balance sheets and income statements, along with more than 1.1 million supply links and over 25,000 bank-firm loans. A unique feature of this model is that the 1:1 representation of the real economy is linked to financial system through credit exposures. It allows us to study how risks spread from the real economy (production layer) to the financial layer. We follow the philosophy that parameters in ABMs should not be freely assumed but fully are determined and fixed by the data. This is the reason why we refrain from including additional features to the model that we cannot calibrate with available data, such as details of the rewiring dynamics. For estimating the production functions of the companies and a simple supplier replaceability dynamics we follow Diem et al. (2022a); Diem et al. (2024a). As, at this stage, we do not employ rewiring dynamics the model is not yet predictive or generative. It has been conceived in the spirit of stress testing. It is designed to estimate and monitor the likely costs of hypothetical external shocks, assuming that there are no interventions of governments or central banks. It is not designed for medium- and long term predictions but rather for short time scales below the typical times that are needed to restructure the SCN.

Within this framework we analysed three central aspects of how financial stability is affected by SCN contagion: (1) Do individual firms pose a threat to financial stability? (2) How is credit risk of banks amplified by supply chain contagion? (3) Do real economy crises such as COVID-19 threaten financial stability?

Our results show that the failure of very few individual firms (those that can cause large cascades of supply chain contagion), can lead to system-wide bank equity losses in the banking sector of around 22% (assuming an LGD of 100%). These losses can be mostly attributed to supply chain contagion induced defaults of firms (indirect losses); the loan write-offs caused by the initially failing firms (direct losses) only play a minor role. Approximately 20% out of the 22% of losses, can be linked to firms defaulting only due to their liquidity buffers being insufficient to withstand the indirect losses from supply chain contagion. This implies that increasing liquidity provision to firms that do not face negative equity levels can alleviate most of the loan defaults that are caused by the failure of high FSRI firms and the associated supply chain contagion.

Our results also indicate, that those firms causing the largest losses do have no, or relatively small, loans from Hungarian banks (they could either borrow from the capital market or from banks abroad), while firms with the largest loans cause small indirect supply-contagion-induced financial losses. This implies that firms with the largest bank-loans might not be the most relevant for monitoring financial stability, but those that can cause substantial indirect defaults. This type of firms would be clearly missed when not taking supply chain contagion into account. It is indicated that regulators could benefit from building up capabilities to monitor supply chain generated and amplified systemic risks. Governments and regulators could collect this type of data as done today already in about a dozen of countries (Bacilieri et al., 2022; Pichler et al., 2023).

Within our framework we performed a stylized stress test by sampling initial shocks based on randomly failing about 1% of firms (based on their estimated default probabilities) and simulating the spread of the production network contagion to the financial layer. In this way we create 10,000 initial shock scenarios and obtain two loss distributions for each bank, one with and the other without SCN contagion. The results suggest a substantial amplification of financial risks, when SCN contagion is taken into account. On average, expected losses of banks are amplified by a factor of 5.2, value at risk — by a factor of 6.7, and expected shortfall — by a factor of 4.4. Note that this large amplification factors are driven by the failure of high systemic risk firms causing an extremely fat tail of the loss distribution. The median amplification factor for the system wide losses is close to 2. Amplification factors differ across banks. If initial failures trigger supply-chain contagion, then direct losses of bank equity are negligible in comparison to the indirect losses caused by SCN contagion. This implies that shock propagation in SCNs can potentially lead to substantially fatter tails in banks' loan defaults, suggesting that capital ratios should factor in these risks.

The observed amplification of credit risk measures by factors between 4.4 and 6.6 unambiguously demonstrates that supply chain contagion is an extremely important factor for a realistic credit risk assessment in a world with increasing numbers of supply chain disruptions. This suggests that current practice could systematically and grossly underestimates capital requirements and sustainable interest rates when large exogenous shocks trigger SCN contagion. The present study strongly indicates that supply chain contagion poses unexpectedly large systemic risks to financial stability. In practice the default probability estimates (performed by banks) implicitly also account for defaults due to SCN contagion occurring in normal times. This is because the data (past default events in the loan portfolios) contains defaults that originated from SCN contagion, (e.g., the failure of a major customer or supplier). On the one hand, our results hint at the fact that SCN contagion actually is a substantial driver of credit risk. On the other hand most data points correspond to times where no major SCN contagion has occurred or it has been cushioned by government interventions, such as during the COVID-19 pandemic. Therefore, credit risk might be underestimated. Mis-estimations of correlations in housing loans played a major role in causing the financial crises 2007/2008. In the similar manner, ignoring that supply chain networks have the potential to cause substantial default correlations can lead to unexpected financial losses due to neglecting economic mechanisms leading to correlated defaults — this time of firms. A large research effort on how the interconnections of financial networks led to financial contagion (Gai and Kapadia, 2010; Glasserman and Young, 2016; Martinez-Jaramillo et al., 2019 among others) led to higher levels of interbank crisis preparedness. The relatively mild progression of the 2023 banking system turmoils (Silicon Valley Bank default, Credit Suisse take over, etc.) demonstrates that the knowledge and experience gained helped in managing risks for financial stability of this crisis effectively. Similar lessons learned are needed for supply chain contagion.

Our framework provides a number of insights into how supply chain contagion can affect financial stability during crisis, unveiling mechanisms, so far hidden from most policy makers. Our simulations show that a COVID-19 shock (calibrated from firm-level employment data) leads to defaults that are predominantly caused by illiquidity of firms. The direct system wide financial losses amount to 3% of total bank equity while indirect losses due to supply chain contagion to 3.2%. This is an amplification of bank losses by more than 100% due to supply chain contagion (amplification factor 2.05). Note that this amplification factor is similar to the ones found for total output losses after e.g., Hurricane Katrina (Hallegatte, 2008), or the first UK lockdown (Pichler et al., 2022b). To assess how slightly different real economy shocks to firms – that are of the same aggregate size, but affect individual firms differently – would affect financial stability we simulate the SCN contagion for 1000 synthetic COVID-19 shocks. Here the losses range from 4.7% to 10.4% of total bank equity. It is important to realize that even though the simulated real economy shocks are of the same aggregate size they cause different losses to the financial system. Hence, it is crucial to model SCN contagion at the firm-level to estimate the effects on financial stability realistically, as has been argued already for assessing real economy shocks Diem et al. (2024a). The firm-level shock propagation framework presented here allows us to calculate liquidity and equity shortages for each individual firm for a given crisis scenario and hence allows the design of targeted policy interventions. By providing liquidity support to firms in an optimal way 80% of the financial system losses can be avoided with a liquidity support budget that is amounting to 0.05% of total bank equity. An in-discriminant approach would cost more than double, but reducing equity losses by merely an additional 0.3% Our findings obtained from the COVID-19 inspired contagion example endorse the widespread practice of providing liquidity support to firms during crises. Yet, by gaining in-depth knowledge about the SCN contagion-caused losses of individual firms, financing support can be tailored in the most effective manner, while curbing the detrimental effects of the crisis and keeping inflation under control.

The presented study has a number of obvious limitations. Foremost, our results depend on a range of assumptions that are relevant for the micro-simulation model. These include the supply-chain contagion mechanism itself, how production disruptions due to supply-chain contagion translate into financial losses for firms, and how the insolvency of firms translates into capital losses for banks. We take a pragmatic approach for firm behaviour. In the proposed setting we assume that revenues and material costs are reduced proportional to the production loss that a firm suffers after the SCN contagion. In this way all firms are treated equally without any consideration of their industry affiliation, or nuances in their business models. Given the present data it is not feasible to account for varying transmission mechanisms according to business models. However, future improvements could take into account the industry sector of firms and other not yet utilized items of their income statements and balance sheets. A second assumption is the condition under which firms default. We use firm's equity and short term liquidity (cash, short term claims and securities subtracted by accounts payable as buffers against reductions of firm profits (profit can turn negative). If the losses exceed one of the buffers we use this as an indicator of default. Our current buffer calculation neglects that firms could, e.g., ask for additional liquidity from banks, or tap additional equity from their owners and, hence, avert a default. However, we provide a sensitivity analysis of losses if liquidity and equity buffers of firms would be up to a factor of two higher. The design of the model fits the purpose of investigating network importance of a firm, nevertheless, does not make the method predictive. In the COVID shock example financing support is provided post factum and is not incorporated into the contagion. A further assumption is the value of the loss given default (LGD). On practice observed LGDs are typically lower than 100%, but since our results are scalable with LGD, we can choose 100% and effects from lower values is than straightforward to infer. Thus, we do not take into account the possibility of instalment postponement, restructuring of firms in case of any problems or use of collateral by banks. However, for the short term view this is not unrealistic as collection and resolution processes can last years. Further, the production loss of a firm, its bankruptcy and default, and losses of banks happen immediately, without a detailed concept of time being modelled. In this manner our results suggest an upper limit of possible losses. Finally, the data even though unique in coverage has shortcomings. The supply-chain network is not entirely complete; small firms and import–export links are missing. Additionally, we exclude firms with missing or atypical (negative) financial variables. We only included the largest banks due to data availability, which, however, should affect results only to a limited extent since the majority of bank-firm loans and loan volume are covered. Note that for our baseline modelling choices on loss given default, the supply chain contagion mechanisms, and the assumption that firms have no access to additional equity or liquidity, our base-line results provide rather upper bounds of financial system wide losses. By conducting sensitivity checks on these model parameters we showed how the losses would change, when these assumptions change.

Data driven 1:1 models along the lines presented here can be used as a basis for assessing governmental recovery and rescue policies, such as loan guarantees for firms, revenue loss subsidies that have been implemented, for example during the COVID-19 pandemic. Applying it to specific shock scenarios such as devaluations of assets due to climate change, would be natural future extensions. Model assumptions would have to be calibrated to the specific initial shock scenario. Maybe the most important step to make the model more practicable, also as a predictive tool, is to include price information that is so-far missing. Further, the contagion between banks needs to be modelled as well as the financing for firms' production and expansion activities. Also more detailed behavioural mechanisms for firms and banks should be included in future work.

In a wider context, the demonstrated risk amplification mechanism through the coupling of the financial network to the SCN can be seen as a concrete example that "networks of networks" are generally expected to show a significant risk-increase in comparison as defaults on simple networks (Gao et al., 2011).

CRediT authorship contribution statement

Zlata Tabachová: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation. **Christian Diem:** Writing – review & editing, Writing – original draft, Validation, Supervision, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **András Borsos:** Resources, Investigation, Data curation. **Csaba Burger:** Resources, Data curation. **Stefan Thurner:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Conceptualization.

Funding sources

This work was supported by the Anniversary Fund of the OeNB (grant no. 18696); FWF Austria Science Fund (grant no. 10.55776/I5985) and Federal Ministry for Climate Action, Environment, Energy, Mobility, Innovation and Technology (BMK) (grant no. GZ 2021-0.664.668).

Appendix A. Economic systemic risk index

We use the supply-chain network, W, ($n \times n$ matrix, $n = 243,339$), where W_{ij} is the trade volume between buyer, j, and supplier, i. Total purchase volume of firm i, the *in-strength*, is $s_i^{\text{in}} = \sum_{j=1}^{n} W_{ji}$, and total amount of sales, the *out-strength*, is $s_i^{\text{out}} = \sum_{j=1}^{n} W_{ij}$. Additionally, we compare these values with revenues and costs from the financial statements of firm. If the out-(in-) strength is smaller than revenue (costs) of a firm, we take the higher value. In such a way we slightly adjust firms' outputs and inputs for exports and imports that are, so far, missing in the SCN.

Every firm, i, is equipped with a generalized Leontief production function, defined as

$$x_i = \min\left[\min_{k\in\mathcal{I}_i^{\mathrm{es}}}\left[\frac{1}{\alpha_{ik}}\Pi_{ik}\right],\ \beta_i + \frac{1}{\alpha_i}\sum_{k\in\mathcal{I}_i^{\mathrm{ne}}}\Pi_{ik},\ \frac{1}{\alpha_{l_i}}l_i,\ \frac{1}{\alpha_{c_i}}c_i\right]. \tag{A.1}$$

Π_{ik} is amount of the input of type k of firm i, $\mathcal{I}_i^{\mathrm{es}}$ is the set of essential inputs and $\mathcal{I}_i^{\mathrm{ne}}$ is the set of non-essential inputs of firm i. Vector p assigns NACE4 industry classification (615 industries) to firms, which also indicates the product type which company supplies. The parameters α_{ik} are technologically determined coefficients, β_i is the production level possible without non-essential inputs $k\in\mathcal{I}_i^{\mathrm{ne}}$ and α_i is chosen to interpolate between the full production level (with all inputs) and β_i. All parameters are determined by W, $\mathcal{I}_i^{\mathrm{es}}$ and $\mathcal{I}_i^{\mathrm{ne}}$. Note, that labour, l_i and capital, c_i, are not explicitly modelled, due to the short-term perspective of our model. The initial labour shock is modelled as the remaining production level, ψ_i.

We simulate how the initial supply and demand reductions spread downstream to the direct and indirect buyers and upstream to the direct and indirect suppliers of the initially affected firm, by recursively updating the production levels of all firms in the network. We update for each firm i the production output at $t+1$, given production levels of its suppliers, $h_j^{\mathrm{d}}(t)$, at time t as

$$x_i^{\mathrm{d}}(t+1) = \min\left[\min_{k\in\mathcal{I}_i^{\mathrm{es}}}\left(\frac{1}{\alpha_{ik}}\sum_{j=1}^{n}W_{ji}h_j^{\mathrm{d}}(t)\delta_{p_j,k}\right),\right. \tag{A.2}$$
$$\left.\beta_i + \frac{1}{\alpha_i}\sum_{k\in\mathcal{I}_i^{\mathrm{ne}}}\sum_{j=1}^{n}W_{ji}h_j^{\mathrm{d}}(t)\delta_{p_j,k},\ \psi_i x_i(0)\right].$$

The production output of firm i at $t+1$, given the production level of its customers, $h_l^{\mathrm{u}}(t)$, at time t is computed as

$$x_i^{\mathrm{u}}(t+1) = \min\left[\sum_{l=1}^{n}W_{il}h_l^{\mathrm{u}}(t),\ \psi_i x_i(0)\right]. \tag{A.3}$$

The economic production losses depend on the choice of production functions we assign to firms. Leontief production functions, where the minimal input of the essential inputs defines the output, makes firms more sensitive to supply chain disruptions; the absence of at least one essential product will lead to the termination of the entire production. Linear production functions, on the other side, are less sensitive to missing inputs, since the output is reduced proportionally to the reduced input only. Here we use a carefully calibrated generalized Leontief production function that distinguishes between firms' essential inputs (treated in the Leontief way) and firms non-essential inputs (treated in a linear way).

Note that due to a lack of data, we cannot consider imports and exports directly as the data does not allow for this. However, we implicitly take imports and exports into account by comparing the in- and out-strength of firms with their revenues and material costs. Then, if a firm has a revenue of 100, but its out-strength (sales within Hungary) are 80, we would adjust for this discrepancy. As fully losing a customer buying goods worth 20, would be based on the out-strength wrongly assumed to have a 25% impact of a firms output, we correct with the revenue such that the actual impact is 20% (the customers actual share of the firms sales). Once import/export data is available we can consider it also explicitly in our model set up, by extending the supply chain network with suppliers and buyers abroad. More on the impact of production functions on supply chain contagion is found in Diem et al. (2022a, 2024a).

The algorithm also contains variable σ which allows for substitution of firms defined as

$$\sigma_i(t) = \min\left(\frac{s_i^{\mathrm{out}}(0)}{\sum_{j=1}^{n}s_j^{\mathrm{out}}(0)h_j^{d}(t)\delta_{p_j,p_i}},1\right) \tag{A.4}$$

It is based on market share of firms with respect to their outputs. Companies with big market share are harder to replace than companies with a small share. For details and the full algorithm, see the methods section in Diem et al. (2022a).

The algorithm can be used to assess a systemic importance of every firm in the SCN. By defaulting a firm, i.e., using a scenario $\psi^{S,j}$, and running the supply chain contagion model, we end up with remaining production levels at time T of firms in the network represented by a vector, $h(T)\in[0,1]^n$. The *economic systemic risk index (ESRI)* of a firm j is now defined as

$$\mathrm{ESRI}_j \equiv \sum_{i=1}^{n}\frac{s_i^{\mathrm{out}}}{\sum_{l=1}^{n}s_l^{\mathrm{out}}}(1-h(\psi^{S,j},T)) \tag{A.5}$$

In Fig. A.9(a) we show ESRI of firms (purple dots) in decreasing order, which reveals characteristic shape (high-risk plateau, steep (power-law) drop, and many small-risk firms) as observed in Diem et al. (2022a). These features obviously carry over to the FSRI values (blue circles) as shown in Fig. 2(a). The 24 most systemically risky companies lead to production losses between 10% and 32% in case of their failure. These firms coincide with the high-risk firms in the FSRI plateau. This is visible in the log–log (plus zero section) scatter plot in Fig. A.9(b) with ESRI on the x-axis and FSRI on the y-axis. Bubble size is again proportional to out strength. We see that for the highest values of ESRI and FSRI are strongly correlated. The overall correlation coefficient of non-zero values is about 0.5 on the log–log scale.

There are many firms with an FSRI equal to 0, but relatively large ESRI values up to around 0.5%. The likely reason is that firms' supply-chain cascades, even though large, do not affect the financial conditions of other firms severely enough to cause their default or cause defaults to firms without loans. Again, many firms do not have bank loans; see Section 3.

Appendix B. Variables in the model

Within the scope of multiple assumptions that frame our model we have two degrees of freedom, variation of which can change our results quantitatively but not qualitatively. These two are definition of default and the loss given default (LGD) parameter κ.

B.1. Definition of default

As introduced in the Method section and discussed throughout the Results, we examine two types of buffers firms can use to cover profit losses from SC disruptions: equity and liquidity. Unlike with firm's equity there is more variability in choice of liquidity. Opting just for firm's cash position would most likely underestimate its real liquidity, given the presence of other liquid short term assets. On the other side, firm would probably use its liquid assets to cover some of its short term liabilities. We assume that firm's available liquidity in short term would be restricted by short term liabilities for goods and services, and, thus, we define liquidity as the difference between short term assets and short term liabilities for goods and services. Alternative choice of liquidity buffer, for example, would be just short term assets without subtraction of liabilities. In Fig. B.10(a) we show financial systemic risk indices of firms (pink stars) for this default buffer. Systemic risks are smaller, because the buffer is higher and absorbs more losses.

In accordance with the Insolvency Code (Section 27 (2f, 2a)) insolvency ($z_i<0$) or illiquidity ($(a_i-s_i)<0$) of a firm can lead to its bankruptcy and, thus, unlikeliness to repay loans. Given this choice, we must assume that before the shock, at time t_0 all firms in our dataset have non-negative and existing equity, liquidity and profit variables. In other words, default indicator vector, $\chi(t_0)$, defined by (2) must be a zero vector. To satisfy the latter, we filter out firms that, based on their financial data, do not meet these conditions. As a result, we exclude 102,107 out of which 6653 are borrowers holding 25% of loan volume equal to 31% of banking system equity (total equity of 27 banks). Initial total volume of loans against banks' equity is equal to 1.27% and after the data filtration the ratio is equal to 0.96.

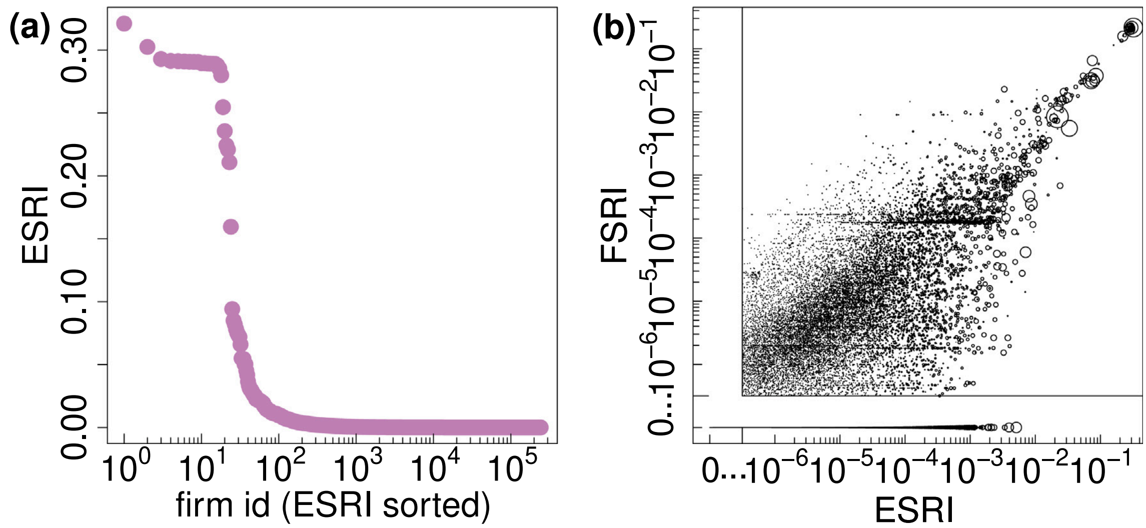

Fig. A.9. Comparison of the economic systemic risk index (ESRI) with FSRI. (a) log-linear plot of firms' ESRI (purple dots) rank-sorted in decreasing order from 1 to 243,339. The ESRI of a firm is the fraction of lost production in the supply chain network after an initial disruption of that particular firm and the ensuing shock propagation. The distribution of ESRI is qualitatively very similar to FSRI. The 24 most systemically risky companies lead to production losses between 10% and 32%. These are mostly the same companies appearing in the FSRI profile in Fig. 2. This is evident also in the scatter plot in panel (b). Bubble size is the out strength (sales). Most risky firms have high out strength, but there are firms with small out strength and high systemic financial and/or economic risks. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

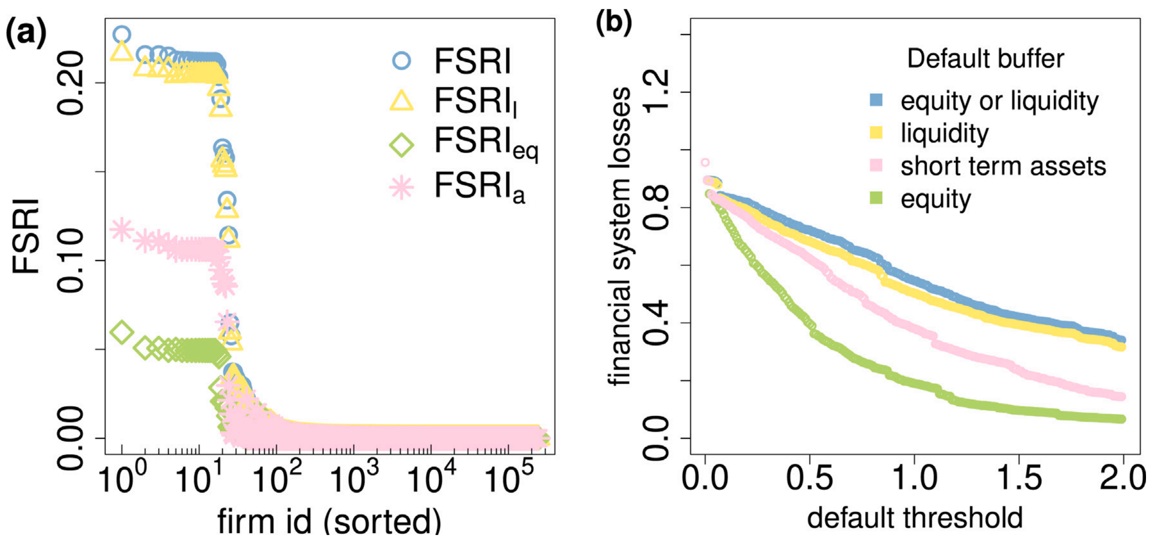

Fig. B.10. FSRI with different buffers and upper possible losses dependent on the threshold parameter. (a) Log-linear scatter plot of FSRI of firms sorted in decreasing order of FSRI. The x-axis refers to firms' FSRI-rank from 1 to 243, 339, the y-axis represents the FSRI values of those firms. FSRI is shown for two causes of defaults independently: FSRI_a - due to short-term assets FSRI_l (yellow triangles) - due to short-term liquidity and FSRI_eq (green diamonds) due to equity insufficiency. FSRI (blue circles) is due to either one of them. All distributions are sorted in the same way with respect to FSRI. Out of 243, 339 firms in the SCN about 100 are systemically risky with 24 (first 24 from left) having very large systemic risk, 10-22% of the overall banks' equity is lost if any of them fails. These companies are the same for all four distributions. Height of plateau differs across different buffer choices. Panel (b) shows upper possible system financial losses for different default buffers (equity, liquidity, short term assets and equity either liquidity) for default threshold $\theta \in [0, 2]$ (we skip the subscript notation). The left-most point refers to a scenario in which all firms with loans default and banks suffer maximum of possible losses equal to 0.96. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

The default indicator definition (2) can be rewritten in the following way

$$\chi_i(\psi) = \begin{cases} 1 & \text{if} \quad \frac{\Delta p_i}{z_i} \geq \theta_{\text{eq}} \quad \text{or} \quad \frac{\Delta p_i}{a_i - s_i} \geq \theta_{\text{l}} \quad , \\ 0 & \text{if} \quad \frac{\Delta p_i}{z_i} < \theta_{\text{eq}} \quad \text{and} \quad \frac{\Delta p_i}{a_i - s_i} < \theta_{\text{l}} \quad . \end{cases} \tag{B.1}$$

where $\theta_{\text{eq}} = \theta_{\text{l}} = 1$. We added Δp_i to the both sides and divided them by $z_i > 0$ or $(a_i - s_i) > 0$. Newly introduced parameters θ_{eq} and θ_{l} are *default thresholds*. $\theta_{\text{eq}} = \theta_{\text{l}} = 1$ means that a firm does not default until profit losses do not exceed 100% of equity or liquidity. Nevertheless, losses around 90% of equity can also indicate upcoming default, and other values of θ_{eq} or θ_{l}, especially slightly below 1, are also possible.

Upper possible financial losses for different default buffers (equity, liquidity, short term assets and equity either liquidity) and default threshold $\theta \in [0, 2]$ (we skip the subscript notation) are visualized in Fig. B.10(b). These losses refelect a hypothetical extreme scenario in which every firm would entirely stop its production and default on its loans, i.e. $h_i(T) = 0$ in (1) and, thus, $\Delta p_i = r_i - c_i$ for every i. This gives the maximum losses the banking system can suffer from commercial loan defaults. One can see, that choice of θ influences results in a big extent. Upper losses for $\theta = 1$ and defaults given by equity either liquidity, liquidity, short term assets, equity are 55%, 50%, 38%, 19% respectively. Since all the previous results are quantified with $\theta = 1$, we can compare with the plateau levels of FSRI distributions respectively

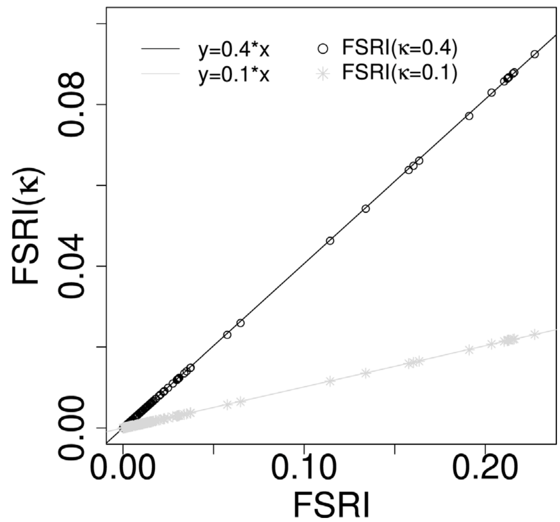

Fig. B.11. Fit of FSRI against FSRI(κ) FSRI = FSRI(κ = 1) of all 243,339 firms is on x-axis. On y-axis are values of FSRI(κ) for κ = 0.4 (black circles) and κ = 0.1 (grey stars). Two lines show linear fits with estimated coefficients 0.4 and 0.1, as expected. The coefficient of determination, r^2, is equal to 1 in both cases, which means that modelled values exactly match observed ones.

at 21%, 20%, 11%, 5% see Fig. B.10(b) Situation in which all loans default is at $\theta = 0$, where results for all default buffers overlap and are equal to 0.96. As mentioned before, this number is equal to the total loan volume against banks' equities.

B.2. Loss given default

As discussed in the Method section, we set LGD indicator κ constant for all firm-bank loans and equal to 1 (entire outstanding principal is written off by a bank). In such a case κ can be factorized out in (4), and definition of FSRI, (7), can be rewritten as

$$\begin{aligned} \mathrm{FSRI}_j &= \sum_{k=1}^{m} \frac{e_k}{\sum_{\ell=1}^{m} e_\ell} \min\left[1, \kappa \sum_{j=1}^{n} \chi_j \frac{B_{jk}}{e_k}\right] \\ &= \kappa \sum_{k\in M} \frac{e_k}{\sum_{\ell=1}^{m} e_\ell} \sum_{j=1}^{n} \chi_j \frac{B_{jk}}{e_k} + \sum_{k\in \tilde{M}} \frac{e_k}{\sum_{\ell=1}^{m} e_\ell} \quad , \end{aligned} \tag{B.2}$$

where M and $\tilde{M}$ are index sets of banks which do not and do have losses exceeding their equity size respectively in a given shock scenario. The second summand is not negligible in general and its maximal value depends on the dataset. For reasonable initial shocks $\tilde{M} = \emptyset$. In calculations of FSRI $\tilde{M} \neq \emptyset$ only when initial shock is given by failure of one of the plateau firms. The second summand, in these cases, is of order 10^{-3}, which does not change results substantially. Thus, FSRI is linear in κ for almost all firms in our dataset (for plateau firms is approximately linear), i.e. FSRI($\kappa = \alpha$) = αFSRI(κ = 1). In Fig. B.11 we present a linear fit of FSRI(κ = 1) against FSRI(κ = 0.1) and FSRI(κ = 0.4) of all 243,339 firms. The coefficient of determination, r^2, is equal to 1 in both cases, which means that modelled values exactly match observed ones.

Results given by means of loss distributions are linear in κ if it is equal for all loans. Indeed from (4)

$$\mathcal{L}_k(\psi^{\mathrm{M},\ell}, \kappa) = \kappa \sum_{j=1}^{n} \chi(\psi^{\mathrm{M},\ell}) \frac{B_{jk}}{e_k} \qquad \text{and}$$

$$\mathcal{L}_k^{\mathrm{dir}}(\psi^{\mathrm{M},\ell}, \kappa) = \kappa \sum_{j=1}^{n} \chi^{\mathrm{dir}}(\psi^{\mathrm{M},\ell}) \frac{B_{jk}}{e_k}. \tag{B.3}$$

It follows that amplification factors and their averages are independent of the variable κ, given it is equal for all loans. Risk measures EL, VaR and ES change proportionally, i.e. $\mu_k^X(\kappa) = \kappa\mu_k^X$ and $\tilde{\mu}_k^X(\kappa) = \kappa\tilde{\mu}_k^X$. Substitute into (11) cancels κ out.

Appendix C. Central bank methodology to PD-estimation for all Hungarian firms liable to corporate tax

The default probability (PD) estimation originates from the methodology developed by Banai et al. (2016) and refined by Burger et al. (2022). This methodology estimated PD values for SMEs included in the annual Corporate Income Tax database of the Hungarian National Tax and Customs Administration ('NAV Társasági Adó'), and had at least one loan or leasing contract (i.e. they were registered in the Credit Registry (KHR) for corporates). The model was trained using observations from the years between 2007 and 2019. For the purpose of this current paper, the original estimation procedure was simplified. While the model was still trained on companies with loan contracts, the goal of the simplification was to make PD-predictions possible for all Hungarian corporates, not just the ones with loan contracts. Hence, all loan-contract level information was removed from the estimator function.

The original methodology of Banai et al. (2016) considered a company to be in default if its instalment payment was at least 90 days overdue. Burger et al. (2022) refined this step by considering defaults only if at least 99 percent of all loan amounts was in default. One year PD was predicted, which means that the probability of default in the four quarters following an observation date was estimated. In order to harmonize this approach with the data, annual values of the Corporate Tax database were converted to quarterly values. Balance sheet values were linearly interpolated, while income statement positions were divided by four. In the absence of further information on seasonality and similar metrics, this was considered as an acceptable compromise between modelling needs and data availability. All these transformations boiled down to 163 thousand tax-liable, credit-taker, distinct SMEs from 2007 till 2019, represented by 3.4 million quarterly observations. The PD-estimator relied on a set of macroeconomic variables (3-months BUBOR, quarterly changes to the HUF-EUR exchange rate, unemployment levels), loan contract-level variables aggregated to company level (such as foreign currency-denominated loan flag, longest loan maturity, sum of all loan values in HUF) and company-level financial variables. To keep the independent variables with explanatory power only, the LASSO-dimension reduction algorithm was applied. And while the final estimator function was a logistic regression, nonlinear relations were minimal, as an analysis with the help of gradient boosting has shown.

This present analysis required alterations to the methodology described above. First, all industries but financial corporations were kept in the data, which were originally excluded. Second, the loan contract-related explanatory variables of the original PD-estimator were removed, since the aim of the model was to make predictions possible for all corporates, not just for those with loans. Finally, the definition of active companies was expanded: now companies with zero revenues but with positive total assets were kept in the training set. This training data was used for PD-estimation that was prepared with the help of LASSO and a simple logistic regression.

Using the model trained in this way, PDs were estimated for about 143,000 companies liable for Corporate Tax, with or without a loan contract. There used to be, however, a sizeable mass of companies using alternative taxation formats (EVA, KATA, KIVA), in our dataset about 52,000, before the recent overhaul of the corporate taxation system, hence the algorithm described did not generate PDs for them. To overcome this, a separate, simple PD-estimator was developed, on the training data again. It was based on an elementary, but available information, such as industrial affiliation, company age, revenue category, a category based on the number of employees, and finally, legal form. While this simplification is certainly not the most sophisticated, it provides reasonable estimations of PD values. Finally, PDs for all 195,000 companies were generated.

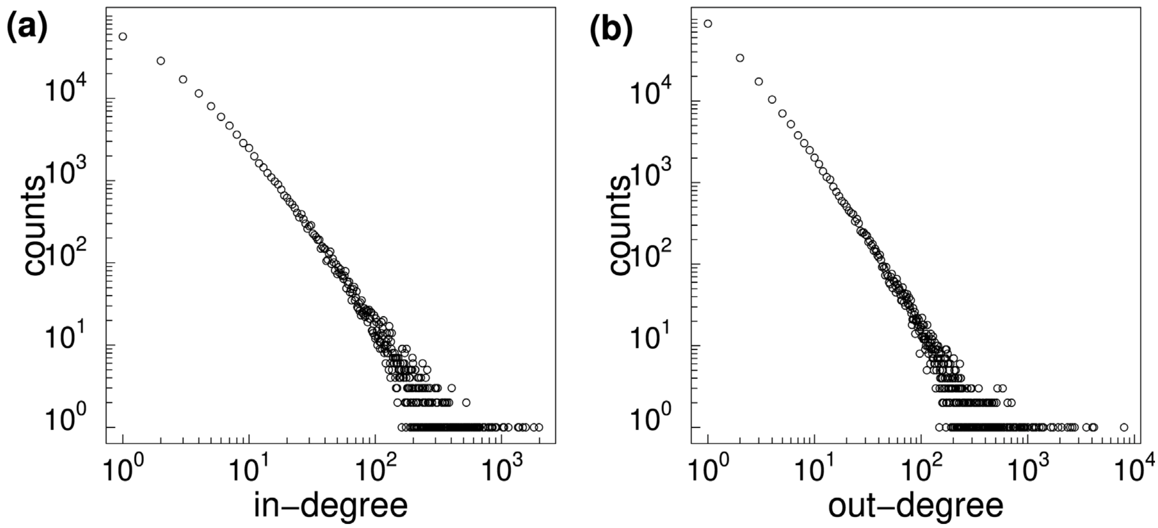

Fig. D.12. Degree distributions of supply chain network W. (a) shows log–log scaled distribution of in-degrees (number of suppliers) of the supply chain network W. There is 81,125 and 56,563 firms with in-degree equal to 0 and 1 respectively. The maximal in-degree in the network is 1981. (b) shows log–log scaled distribution of out-degrees (number of buyers) of the supply chain network W. There is 51,679 and 88,810 firms with out-degree equal to 0 and 1 respectively. Maximal out-degree in the network is 8004. Average in(out)-degree is 4.5.

Table D.3
Quantiles of in, s^{in}, and out, s^{out}, strengths of 243,339 firms from the supply chain network W (in 10^3 2019 HUF).

	Out strength	In strength
Mean	230 424	230 424
std	4 874 972	5 987 263
Skewness	133	134
Kurtosis	25 344	26 094
20%	0	0
30%	2286	0
40%	5563	1785
50%	10 230	5264
60%	17 620	11 631
70%	32 217	24 643
80%	69 346	56 032
90%	200 671	171 115
99%	2 771 296	2 605 843
99.9%	27 046 345	25 532 697
99.99%	142 470 247	220 741 276

Appendix D. Descriptive statistics of data

A detailed set of descriptive statistics about the supply chain network, firms financial variables and bank-firm loans can be found in Tables D.3–D.6. Fig. D.12 shows the in- and out-degree distributions of firms in the supply chain network with in- (out-)degree on the x-axis and the probablity of observising a firm with degree larger than x on the y-axis.

Appendix E. Details on FSRI

Diem et al. (2022a) show that economic systemic risk of a firm is not necessarily proportional to total sales and Diem et al. (2024b) shows a similar results for establishment-level supply networks – the network structure matters. In Fig. E.13(a), we see that this holds also for FSRI. Firms with very small out strength (above the blue line) can carry a substantial systemic risk. Fig. E.13(b) shows FSRI against loans of firms on a log–log scale. As we mention in Section 3, majority of firms in the SCN do not have loans. Nevertheless, they can cause losses to banks indirectly indicated by the column of bubbles on the left hand side of the plot. The firms on the top of the column, upper left corner, are firms with high systemic risk and with no loans in the country. Other firms above the blue line are also the most risky firms from the plateau in Fig. 2. Companies that are on the diagonal cause only direct losses equal to their own loans. All other firms, above and under the diagonal, cause indirect losses as well. Altogether, there is no clear correlation of loan size and FSRI.

In Fig. E.14(a) we show disaggregated to banks FSRIs of the 80 riskiest firms. The bar-plot shows the contributions from Eq. (7), the losses of 10 banks are shown with different colours, the relative compound losses of the remaining 17 banks in grey. Again, since the failure of the top FSRI firms trigger highly similar supply chain cascades, they trigger highly similar financial losses. This is reflected by the fact that banks are affected in a very similar ways by these 17 initial firm-failures. However, in general not always the same banks are affected. Different banks have different business models and different client structures. Some banks have large corporate loan portfolios relative to their equity, others have a stronger focus on retail- and mortgage lending. Further, banks can have a strong focus on certain industry sectors such as agriculture. Therefore, some firms cause supply chain cascades that affect specific banks disproportionately. The log–log scatter plot in Fig. E.14(b) shows the fractions of loans (x-axis) against the fractions of clients (y-axis) of banks in the financial layer (we skip 10 banks with fractions below 0.25%). Bank 3 has the biggest amount of loans; bank 1 has the biggest proportion of clients. The bar-plot reveals that losses are suffered mainly by 10 banks: 1, 2, 3, 4, 9, 10, 12, 13, 14, 18 covering 70% of the entire equity in the financial layer. These banks hold 89% of clients with 90% of loan volume in those banks. Banks 12, 3, 1, 2 cover 49% of total equity, hold 48% of clients having 55% of loaned money in the system.

Appendix F. Contagion-adjusted probability of default

Here we propose an alternative method for assessing contagion-adjusted system wide and bank specific loss distributions. Based on the supply-chain contagion after the initial failure of each single firm – the calculations underlying FSRI – we identify the so-called *critical suppliers and buyers* of every firm, i. These are direct and indirect suppliers and buyers, which in case of their failure would cause a default of firm i. With this information, we can adjust the "idiosyncratic" probability of default of firm i, pd_i by taking the "idiosyncratic" probabilities of default of its critical suppliers and buyers into account. This gives us the *contagion-adjusted probability of default*. q_i ($q_i \geq \text{pd}_i$). Based on these probabilities, q_i, we sample 10,000 Bernoulli random vectors of size equal to number of firms with the success probability, q_i, indicating that

Table D.4
Quantiles of financial variables of firms satisfying preconditions for definition of default (in 10^3 2019 HUF).

	Revenue (r)	Costs (c)	Equity (z)	Liquidity ($a-s$)	sh.t.assets (a)
Mean	655 001	488 903	393 586	208 162	254 020
std	11 829 827	9 933 412	14 939 329	5 221 769	5 561 594
Skewness	126	125	169	137	124
Kurtosis	25 117	23 922	34 928	28 016	22 949
0.01%	0	0	0	0	0
0.1%	0	0	0	0	0
1%	0	0	113	50	112
10%	8872	3094	3540	2569	3216
20%	16 945	7777	6291	4806	6134
30%	26 864	14 296	10 049	7732	9870
40%	40 412	23 668	15 362	11 769	15 015
50%	60 453	37 508	23 276	17 566	22 506
60%	92 477	60 109	36 185	26 506	34 144
70%	147 411	100 280	59 367	41 786	54 178
80%	263 849	186 255	110 742	73 717	96 276
90%	628 886	456 563	285 657	175 126	229 439
99%	7 691 458	5 613 204	3 444 232	2 007 219	2 588 667
99.9%	68 145 095	49 892 616	35 232 170	22 205 676	29 441 143
99.99%	462 941 207	371 521 699	382 868 449	184 408 455	205 130 360

Table D.5
Quantiles of 102,107 excluded firms' financial variables (in 10^3 2019 HUF).

	Revenue (r)	Costs (c)	Equity (z)	Liquidity ($a-s$)	sh.t.assets (a)
Mean	212 779	161 053	195 287	14 841	71 707
std	7 948 156	3 770 365	12 357 935	4 669 988	1 823 686
Skewness	176	71	123	−248	99
Kurtosis	41 498	6558	17 604	74 410	13 860
0.01%	−2561	0	−9 829 239	−38 082 075	0
0.1%	0	0	−951 911	−1 242 628	0
1%	0	0	−100 057	−127 539	0
10%	0	0	−2458	−2698	0
20%	0	0	0	0	0
30%	0	0	0	0	0
40%	0	0	0	0	0
50%	0	0	0	0	0
60%	0	0	0	0	0
70%	2705	4413	0	0	1514
80%	24 012	24 222	2248	236	6988
90%	99 756	95 814	20 896	7531	31 690
99%	1 655 936	1 554 197	776 383	388 383	687 238
99.9%	28 791 589	22 831 217	18 088 002	7 470 977	11 716 868
99.99%	265 265 085	184 244 253	304 043 434	54 301 790	65 598 330

Table D.6
Quantiles of loans of firms weighted by total equity of 27 banks (in 10^3 2019 HUF).

Mean	0.01%	0.1%	1%	10%	20%	30%	40%
$3.62 \cdot 10^{-5}$	10^{-12}	$2 \cdot 10^{-11}$	$0.01 \cdot 10^{-6}$	$0.35 \cdot 10^{-6}$	$0.75 \cdot 10^{-6}$	$1.24 \cdot 10^{-6}$	$1.93 \cdot 10^{-6}$
50%	60%	70%	80%	90%	99%	99.9%	99.99%
$3 \cdot 10^{-6}$	$5 \cdot 10^{-6}$	$8 \cdot 10^{-6}$	$15 \cdot 10^{-6}$	$42 \cdot 10^{-6}$	$586 \cdot 10^{-6}$	$3217 \cdot 10^{-6}$	$11\,538 \cdot 10^{-6}$

firm i failed. Note that before we simulated $\psi_i^{\mathrm{M},\ell} \sim Ber(\mathrm{pd}_i)$ and used this as initial shock of the supply chain contagion algorithm. In this approach the supply-chain contagion channel is implicitly considered by the adjusted PDs, q. We calculate again the losses $\mathcal{L}_k$ and $\mathcal{L}$ to generate loss distributions of every bank separately and of the whole financial sector, see Figs. F.15 and F.17. In Fig. F.16 we present comparison of results from two methods. Risk amplification factors of banking systemic losses (11) in the case of PD-adjusted method for EL, VaR and ES are 7.1, 5.6, and 4.9 respectively. We see, that the second method unnecessarily captures contagion risk in expected losses. The contagion-adjusted loss distribution given by the first method also increases expected losses, but risks given by the systemically important firms are captured by tails as a less probable events, which they indeed are.

Our aim is to incorporate the supply chain (SC) contagion channels leading to defaults, as described in Section 2, into conventional credit risk measure — the probability of default (PD). Note, that information about SC links is not involved in the PD model of the central bank, described in the Appendix C. Nevertheless, we do not intend to develop any new model which would include supply chain links as additional variables, and rather adjust the existing values of PDs. We use a simple observation arising from our method 2, that failures of some firms lead to defaults of another ones via SC contagion channels. Therefore, PD of the defaulted firm should increase proportionally to PD of the failed firm. We assume, that probabilities of failures of firms (i.e. events that start contagion) are equal to their respective PDs estimated by the central bank model.

Let us recall some notation from the main part. In Section 2 we discussed the production network contagion initiated by the failure of a single firm, j, represented by an initial shock, $\psi^{\mathrm{S},j}$. Contagion leads to production losses of some firms in the network and, in some cases, to bankruptcy and default. The binary vector, $\chi = \chi(\psi^{\mathrm{S},j})$ (2), indicates defaulted firms, $\chi_i(\psi^{\mathrm{S},j}) = 1$ means that a failure of a firm j leads to a default of a firm i. If the latter is true, firm j will be called a *critical*

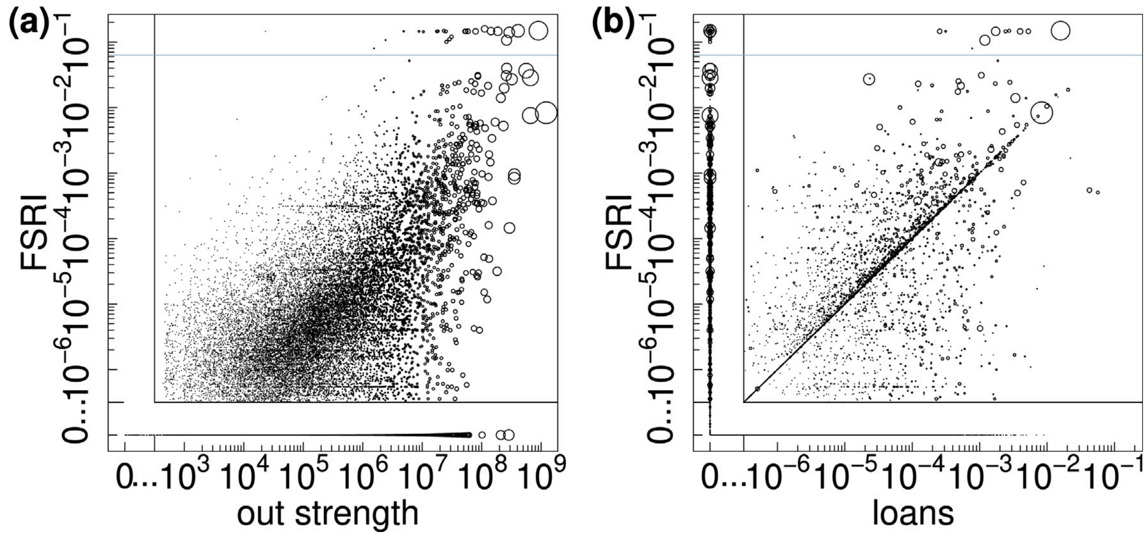

Fig. E.13. FSRI against total sales and loans of firms. The first log–log (plus zero section) scatter plot (a) shows the out strength of firms (x-axis) against their FSRI values (y-axis). Bubble size is proportional to the out strength (total sales) of a company. The horizontal (blue) line is plotted at 0.06 indicating the most risky firms which are above it. (b) shows loans of firms divided by the total equity of 27 banks (x-axis) against FSRI (y-axis). Bubble size corresponds to out strength. Values on the diagonal are equivalent to the direct losses (orange bubbles) plotted in Fig. 3(b).

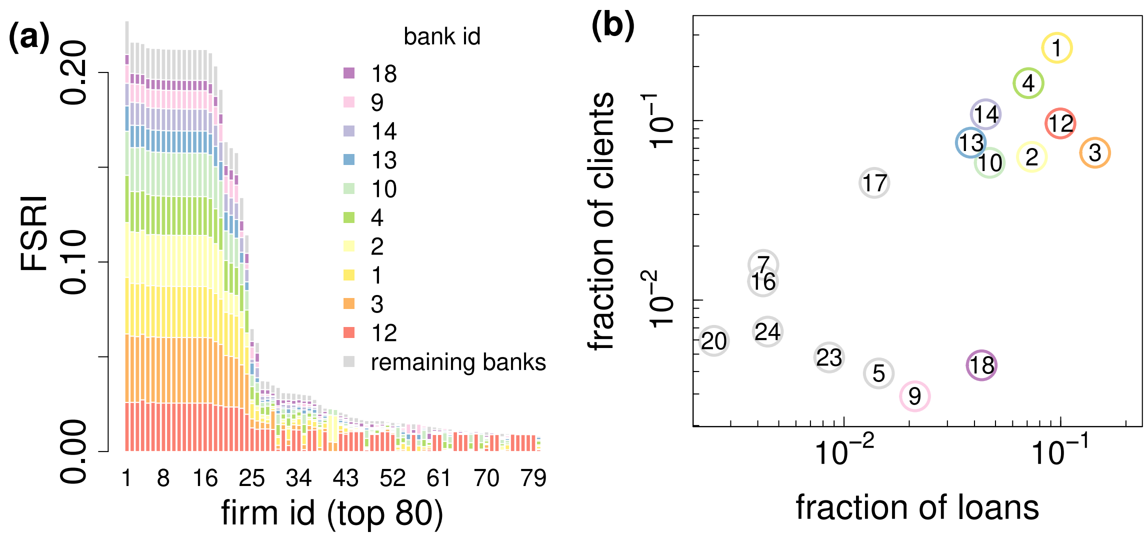

Fig. E.14. FSRI by losses for each bank. (a) FSRI of the first 80 firms disaggregated to banks. Losses are suffered mainly by 10 banks, namely 1, 2, 3, 4, 9, 10, 12, 13, 14, 18. (b) log–log scatter plot shows the fractions of loans versus bank creditors (we omit 10 banks with fractions below 0.25%). It is visible that bank 3 has the largest amount of loans, 17%, and bank 1 has the biggest proportion of clients, 26%. Banks in the upper right corner (1, 2, 3, 4, 10, 12, 13, 14) suffer substantial losses from the riskiest firms in the bar-plot.

supplier or buyer of the firm i. Events where a critical supplier or buyer of a firm defaults are equivalent to events where the firm itself defaults.

Thus, the *contagion-adjusted probability of default*, q_i, of the firm i with $\eta = \eta(i)$ critical links is a joint probability over all events where at least one of the critical buyers and suppliers, or the firm itself, defaults. Let $p_i \equiv \mathrm{pd}_i$ $(i = 1, 2, \dots, \eta(i) + 1)$ denote default probabilities of critical suppliers or buyers of the firm i including the firm itself, then

$$q_i = \sum_{k=1}^{2^{\eta+1}-1} \prod_{l \in I_k} p_l \prod_{j \in I \setminus I_k} (1 - p_j) \quad , \tag{F.1}$$

where I is a an index set with $\eta + 1$ elements, and $I_k \subseteq \mathcal{P}(I) \setminus \emptyset$. $\mathcal{P}(I)$ is the power the index set, with $2^{\eta+1}$ elements. Indeed, in Eq. (F.1) the sum is over all possible combinations of default and non-default of critical nodes, except for the situation when no company defaults.

For example, if a firm $i \equiv a$ has one critical buyer $j \equiv b$ and the independent default probabilities are estimated by the central bank as p_a and p_b, then $\mathcal{P}(I) \equiv \{\{\emptyset\}, \{a\}, \{b\}, \{a, b\}\}$ is the power set with 2^2 elements, $I \equiv \{a, b\}$ and $I_k \subseteq \{\{a\}, \{b\}, \{a, b\}\}$ with $k = 1, 2, 3$. The probability of a default q_a of the firm a conditional on the critical network links can be obtained from Eq. (F.1) as

$$q_a = p_a p_b + p_a(1 - p_b) + (1 - p_a) p_b \quad . \tag{F.2}$$

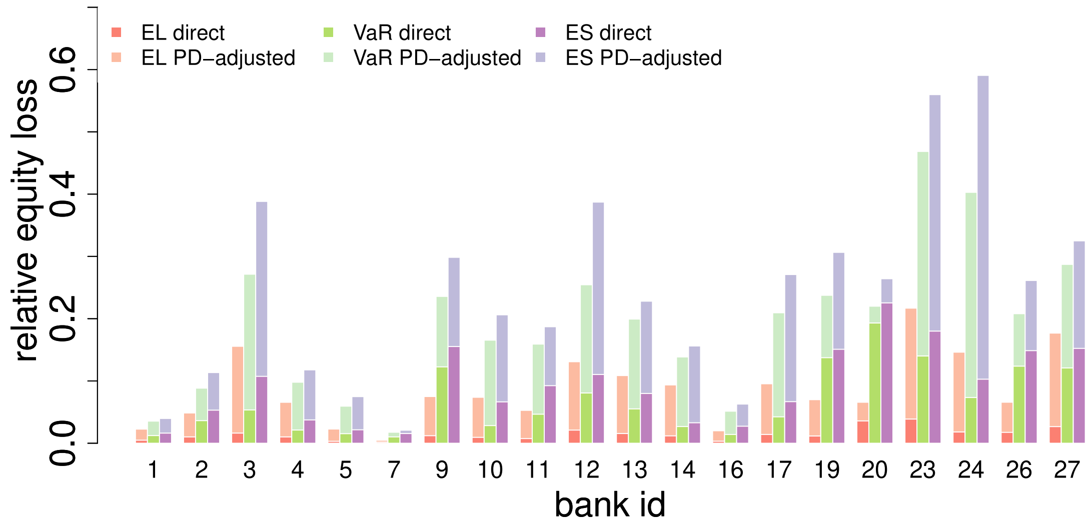

Fig. F.15. Expected loss, value at risk, and expected shortfall of banks from direct and PD-adjusted loss distributions. As Fig. 4, x-axis represents bank indices and y-axis shows risks relative to equity of each bank. We skip banks n.6, 8, 15, 18, 21, 22, 25 which have less than 30 clients. Risk measures EL, VaR, and ES are denoted by red, green and purple colours respectively, for direct and PD-adjusted loss distributions in dark and light shades respectively. The bar-plot is not stacked, meaning that dark bars are always plotted in front of light bars. Average risk amplification factors are $\rho^{\mathrm{EL}} = 5.8$, $\rho^{\mathrm{VaR}} = 3.9$, and $\rho^{\mathrm{ES}} = 2.7$. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

However, we can quantify q_a in a more convenient way. Summation in Eq. (F.1) over $2^{\eta+1}$ elements, including the omitted term,

$$\prod_{j=1}^{\eta+1}(1-p_j) \quad , \tag{F.3}$$

when no firm defaults, is equal to 1. It is because $2^{\eta+1}$ elements span the complete probability space. Therefore, we can rewrite Eq. (F.1) as

$$q_i = 1 - \prod_{j=1}^{\eta+1}(1-p_j) \quad , \tag{F.4}$$

where $q_i \geq p_i$ for every firm i from the supply chain network. The probability of default q_a from the example is thus

$$q_a = 1-(1-p_a)(1-p_b) \quad . \tag{F.5}$$

Finally, we can use the *contagion-adjusted default probabilities*, q_i ($i \in n$), to generate *PD-adjusted loss distributions* of banks and compare them with the contagion-adjusted loss distributions in Figs. 4–5. Again, a Monte Carlo simulation of random defaults sampled from to q_is is performed. In every trial (out of 10,000) losses of a bank are obtained as a sum of all loans of its defaulted clients (i.e. the *loss given default*, LGD = 100%). EL, VaR, and ES (light red, light green and light purple respectively) of PD-adjusted loss distributions in comparison with the same measures of direct loss distributions (Fig. 4) are presented in Fig. F.15. The average risk amplification factors are $\rho^{\mathrm{EL}} = 5.8$, $\rho^{\mathrm{VaR}} = 3.9$, and $\rho^{\mathrm{ES}} = 2.7$.

If we compare these average amplification factors with amplification factors from contagion-adjusted loss distributions presented in Fig. 4, we see, that the ρ^{EL} is higher for the PD-adjustment method, while the ρ^{VaR} and ρ^{ES} are higher in case of the contagion-adjusted approach. Indeed, from Fig. F.16(a) it is clear that ELs from PD-adjusted method are higher or equal for all banks. In the case of VaR and ES, the pattern in Fig. F.16(b)–(c) does not change fully, nevertheless, risks are higher from the contagion-adjusted method for the majority of banks. This means that the contagion-adjusted method can capture SC shocks better, which are less probable, in the tails of loss distributions. Fig. F.17 is an analogue of Fig. 5 with the added PD-adjusted loss distribution (yellow) of the whole banking layer. The x-axis show losses of financial system relative to the overall equity of all banks. Risk amplification factors from Eq. (11) for EL, VaR, and ES of PD-adjusted and direct loss distributions are equal to 7.1, 5.6, and 4.9, respectively.

Appendix G. Details on the COVID-19 shock scenario

Figs. G.18 and G.19 show NACE1 level distributions of direct and indirect production and financial losses from synthetic COVID shock scenaria, $\psi^{\mathrm{C},\ell}$, $\ell = 1, 2, \ldots, 1000$. Synthetic shocks are inspired by the empirical COVID shock, $\psi^{\mathrm{C,emp}}$. They are generated in such a way that aggregated losses within NACE2 (and hence NACE1) sectors are preserved in every scenario, and are equal to aggregated losses in the empirical shock scenario (pink bars in Fig. 7(b)). This is why there is no variance in direct production losses Fig. G.18 (pink dots next to boxplots). The Variance within the direct losses would be visible only on more granular level of individual firms. The biggest initial shocks are suffered by industries C-Manufacturing, D-Energy supply and G-Whole sale and retail trade. Contagion-caused indirect losses show variation across scenarios. This shows that there is heterogeneity in contagion channels which are highly sensitive to initially affected firms. For most sectors indirect losses from contagion are higher than direct losses. The biggest variance is evident within categories affected by the largest direct shocks. Additionally, sectors H-Transportation and storage and J-Information and communication, have, in some cases, almost double indirect losses. More on the heterogeneity of contagion channels in the production networks can be found in Diem et al. (2024a). Fig. G.19 shows box plots of direct (orange) and indirect (blue) financial system losses caused by synthetic COVID shocks for each NACE1 sector. The biggest variance is within sector C-Manufacturing and I-Accommodation and food services and L-Real estate. Direct and indirect financial losses do not necessarily correlate with the respective production losses in Fig. G.18. It is the presence of loans, default buffers and the variatio nof the shock across scenarios that cause the non-linear dependency between production and financial losses across industry sectors.

Fig. G.20 shows risk measures of banks calculated on the losses due to the 1,000 synthetic COVID shock scenaria. Each triplet of bars indicates expected loss (red), value at risk (green) and expected shortfall (purple) of one bank. Dark and light shades show direct risks from initial shock and additional contagion-adjusted risks respectively. In the COVID shock firms suffer partial production losses which do not lead to network contagion of the size when a high systemic risk firm fails. Firms from the high risk plateau (Fig. 2) are never fully affected and they still maintain a certain level of production. Thus, unlike in Fig. 4, here additional losses from the supply chain contagion

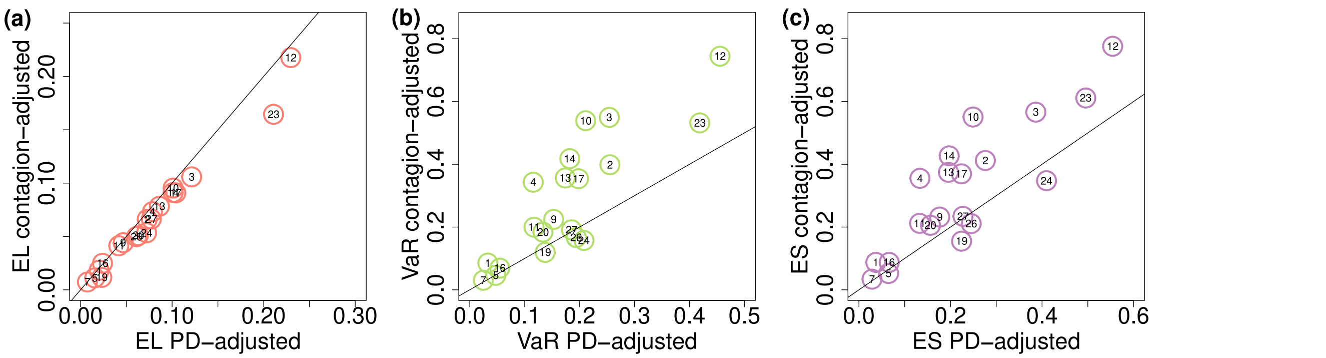

Fig. F.16. Comparison of two methods for the supply chain contagion-stressed loss distributions. Three panels show comparison of the results presented in Fig. 4 (contagion-adjusted method) and Fig. F.15 (PD-adjusted method), for EL, VaR, and ES in (a), (b), and (c), respectively.

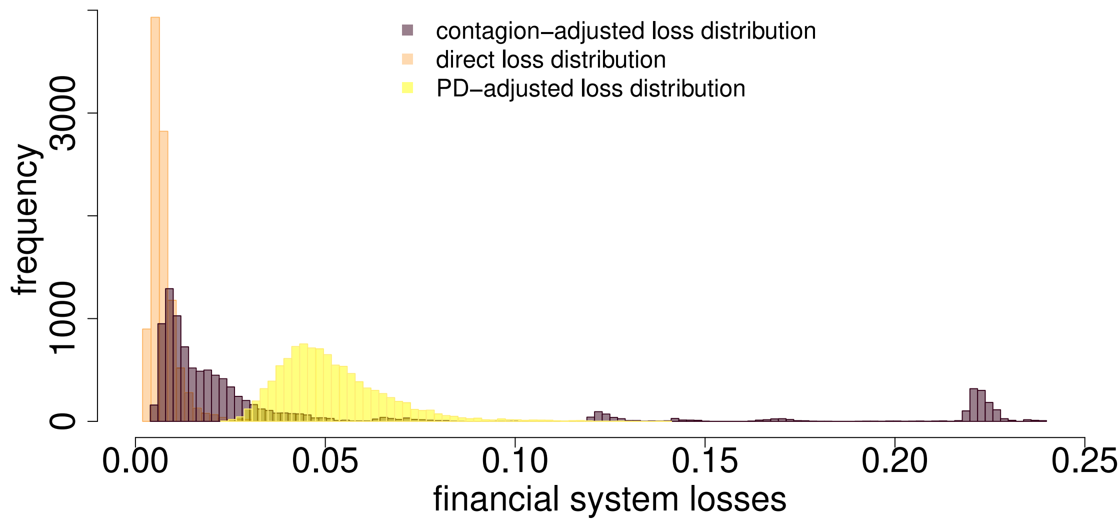

Fig. F.17. Direct, contagion-adjusted and PD-adjusted loss distributions. Distribution of direct losses (orange), contagion-adjusted losses (brown), and PD-adjusted losses across the whole banking layer of 27 banks. The x-axis show losses of financial system relative to the overall equity of all banks, the y-axis is the frequency of losses. The peak around 0.22 corresponds to shock scenarios where firms with the highest FSRI default. Risk amplification factors from Eq. (11) for EL, VaR, and ES of PD-adjusted and direct loss distributions are equal to 7.1, 5.6, and 4.9, respectively. Risk amplification factors for EL, VaR, and ES of contagion-adjusted and direct loss distributions are equal to 6.4, 16.1, and 12.2, respectively. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

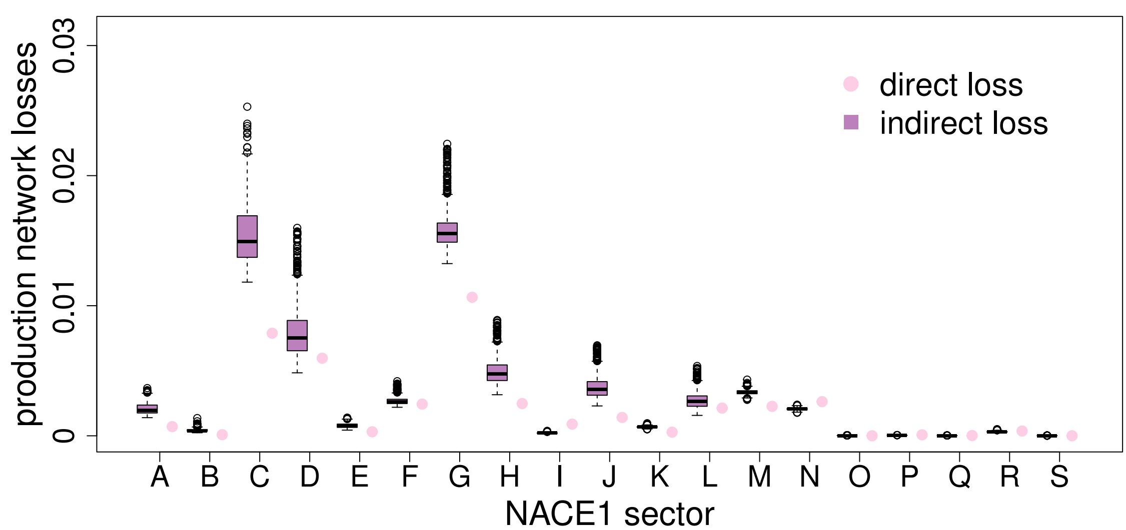

Fig. G.18. Box plots of direct and indirect production network losses initiated by synthetic COVID shocks, $\psi^{C,\ell}$, disaggregated to NACE1 sectors. Letters on the x-axis refer to the top level Nomenclature of Economic Activities (NACE 1). Systemic production losses on the y-axis are equal to output losses given by Eq. (A.5). Pink colour denotes direct losses (caused by initially defaulting firms) and purple colour indicates indirect losses (additional losses from defaults caused by the SCN contagion). Direct aggregated on NACE1 level losses (pink dots) are constant across all the scenaria by definition. The biggest variation of indirect losses is within category C-Manufacturing, D-Energy supply and G-Whole sale and retail trade. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

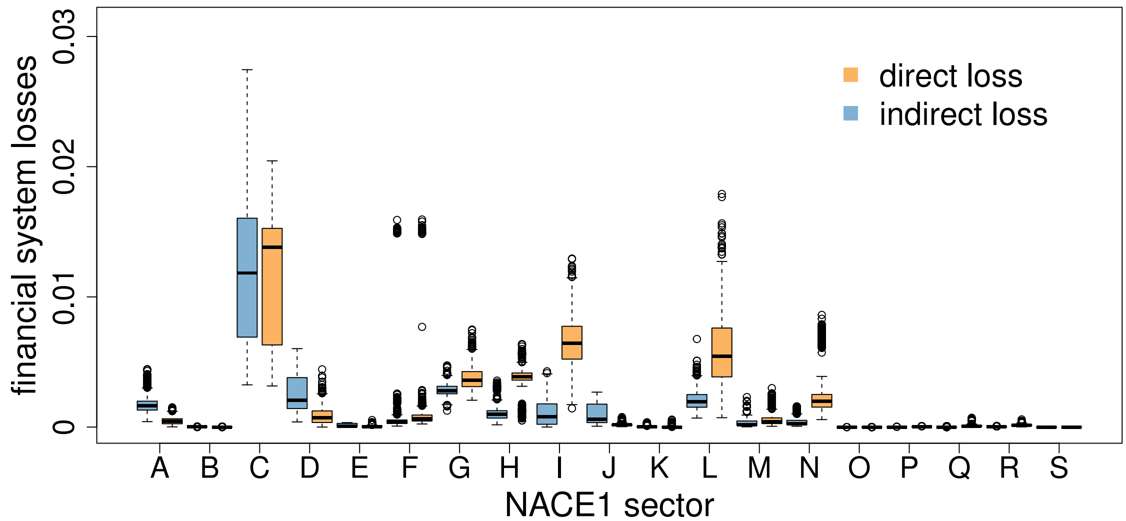

Fig. G.19. Box plots of direct and indirect financial system losses initiated by synthetic COVID shocks disaggregated to NACE1 sectors. Letters on the x-axis refer to the top level Nomenclature of Economic Activities (NACE 1). Systemic equity losses on the y-axis are given by disaggregated $\mathcal{L}\left(\chi(\psi^{C,\ell})\right)$ from Fig. 6. Orange colour denotes direct losses (caused by initially defaulting firms) and blue colour indicates indirect losses (additional losses from defaults caused by the SCN contagion). The biggest variation is within category C-Manufacturing, F-Construction, I-Accommodation and food service activities, L-Real estate activities. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

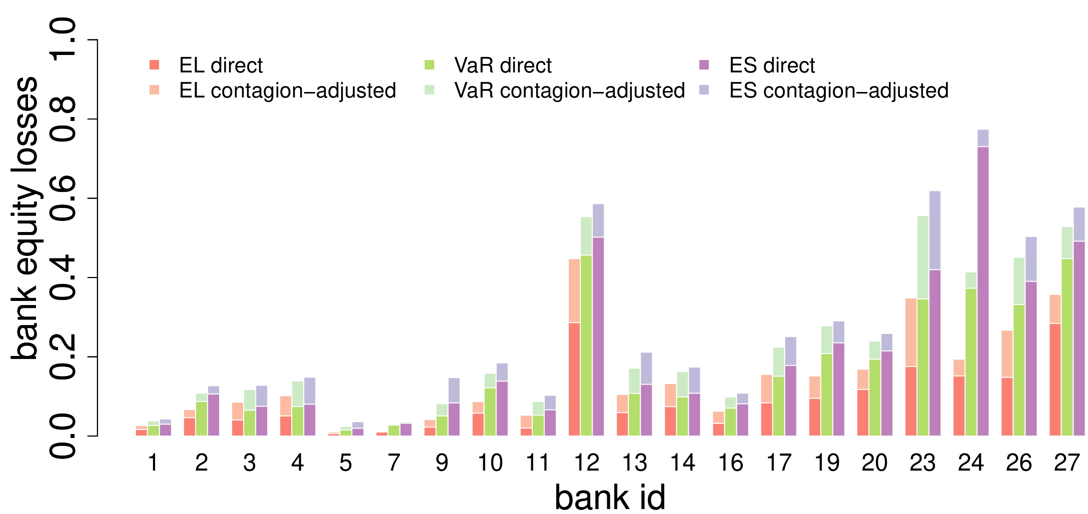

Fig. G.20. Expected loss, value at risk, and expected shortfall of banks from synthetic COVID shocks. x-axis represents bank indices and y-axis shows risks relative to equity of each bank. Banks n.6, 8, 15, 18, 21, 22, 25 which have less than 30 clients are skipped. Risk measures EL, VaR, and ES are denoted by red, green and purple colours respectively, for direct and indirect loss distributions in dark and light shades respectively. The bar-plot is not stacked, meaning that dark bars are always plotted in front of light bars. Average risk amplification factors are $\rho^{\mathrm{EL}} = 1.7$, $\rho^{\mathrm{VaR}} = 1.4$, and $\rho^{\mathrm{ES}} = 1.4$. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

are less extensive. Note that for the COVID-19 shocks also direct losses are higher than for individual firm failures, because the initial shocks affect between 60–80 thousands firms in each scenario. When drawing random shocks based on firms' PDs only around 2500 firms are affected per scenario, but they all suffer 100% production termination. These findings suggest that an supply chain contagion can be larger when a few high systemic risk firms fully fail than when relatively small losses affect a big number of firms.

Appendix H. Effects of liquidity support policies for synthetic covid-19 shocks

Here we show the effects of providing liquidity assistance to illiquid firms for the 1,000 synthetic COVID-19 shocks. Fig. H.21 shows aggregated liquidity support to all illiquid but solvent firms on x-axis, and financial system losses – that would happen if these firms would default – on y-axis. Orange circles in panel (a) indicate results related to defaults from firms being directly affected in each of 1000 synthetic COVID scenaria. Blue circles in panel (b) show results for defaults of firms due to supply chain contagion for each shock scenario. Black crosses in both plots indicate results for direct (a) and indirect (b) shocks in empirical COVID scenario which we described in detail in Section 4.3.2. Circles in both plots are above the diagonal which means that liquidity support to all directly and indirectly illiquid solvent firms in every scenario is less costly than potential losses from their loan write offs. Note that financial system losses on the y-axis are computed with the loss given default equal to 100%. For smaller values of this parameter support can become more expensive. Fig. H.22 is analogical to Fig. H.21, but it shows results for illiquid and insolvent firms. In most of the cases liquidity support to directly affected firms exceeds size of

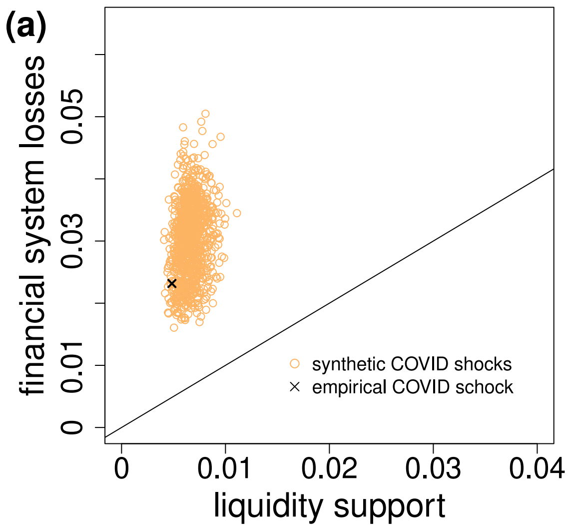

Fig. H.21. Liquidity support for *illiquid solvent* directly and indirectly defaulting firms after the COVID shock. Liquidity support is on x-axes and loan write offs, $\mathcal{L}(\psi^{C})$, are on y-axes of both plots. Circles and x-symbol are referring to 1000 synthetic shocks and empirical shock respectively. In (a) we show results with respect to directly defaulting illiquid solvent firms and in (b) with respect to indirectly defaulting ones. Since all the points are above diagonals liquidity assistance is cost-effective.

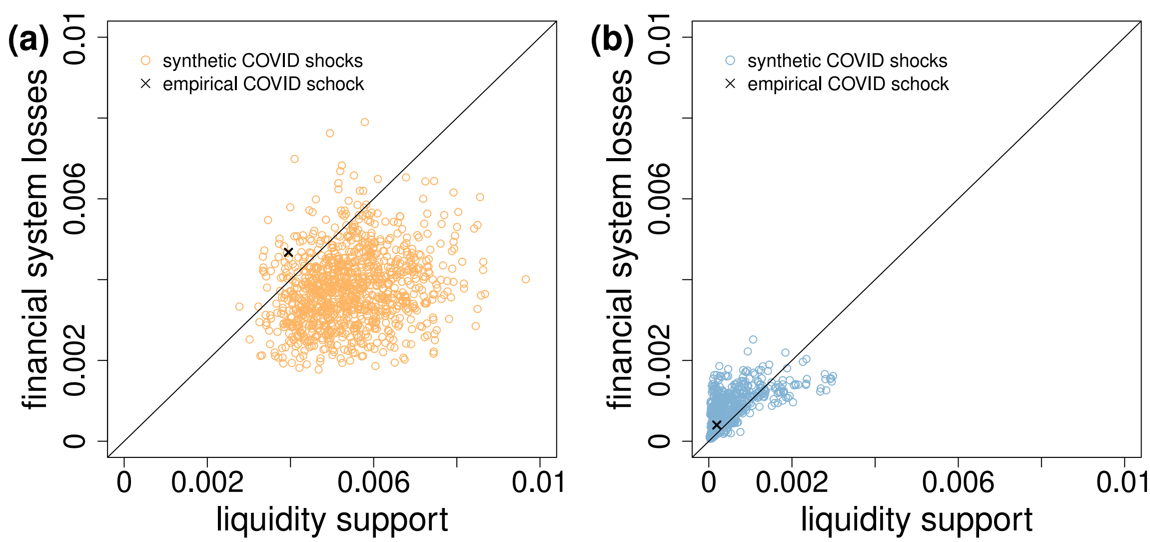

Fig. H.22. Liquidity support for *illiquid insolvent* directly and indirectly defaulting firms after the COVID shock. Liquidity support is on x-axes and loan write offs, $\mathcal{L}(\psi^{C})$, are on y-axes of both plots. Circles and x-symbol are referring to 1000 synthetic shocks and empirical shock respectively. In (a) we show results with respect to directly defaulting illiquid insolvent firms and in (b) with respect to indirectly defaulting ones. In case of direct shocks points (orange) are mostly under the diagonal. This means that liquidity assistance is not cost-effective. Results for indirectly affected firms suggest opposite, nevertheless for smaller LGD final systemic losses might be even smaller and the assistance would become expensive. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

the aggregate losses that could be prevented (orange circles under the diagonal). For indirectly affected firms most of the points are above the diagonal, which shows that additional contagion losses are not so high for these firms. This detailed view on how firms are affected directly and indirectly in each scenario can serve as a useful tool in designing optimal strategies.

References

Acharya, V., Engle, R., Pierret, D., 2014. Testing macroprudential stress tests: the risk of regulatory risk weights. J. Monet. Econ. 65, 36–53. http://dx.doi.org/10.1016/j.jmoneco.2014.04.014.

1991. Hungarian insolvency code (act XLIX of 1991 on bankruptcy proceedings and liquidation proceedings).

Arnold, B., Borio, C., Ellis, L., Moshirian, F., 2012. Systemic risk, macroprudential policy frameworks, monitoring financial systems and the evolution of capital adequacy. J. Bank. Financ. 36 (12), 3125–3132. http://dx.doi.org/10.1016/j.jbankfin.2012.07.023.

Bacilieri, A., Borsos, A., Astudillo-Estevez, P., Lafond, F., 2022. Firm-Level Production Networks: What Do We (Really) Know? Technical Report, work in progress.

Bagehot, W., 1873. Lombard Street: A Description of the Money Market. HS King&Company.

Banai, Á., Körmendi, G., Lang, P., Vágó, N., 2016. A magyar kis-és középvállalati szektor hitelkockázatának modellezése, MNB-tanulmányok 123.

Barrot, J.-N., Sauvagnat, J., 2016. Input specificity and the propagation of idiosyncratic shocks in production networks. Q. J. Econ. 131 (3), 1543–1592. http://dx.doi.org/10.1093/qje/qjw018.

Bartik, A.W., Bertrand, M., Cullen, Z., Glaeser, E.L., Luca, M., Stanton, C., 2020. The impact of COVID-19 on small business outcomes and expectations. Proc. Natl. Acad. Sci. 117 (30), 17656–17666. http://dx.doi.org/10.1073/pnas.2006991117.

Basel Committee on Banking Supervision, 2023. CRE32.6 - Risk components for corporate, sovereign and bank exposures.

Battiston, S., Dafermos, Y., Monasterolo, I., 2021. Climate risks and financial stability. J. Fin. Stab. 54, 100867. http://dx.doi.org/10.1016/j.jfs.2021.100867.

Battiston, S., Martinez-Jaramillo, S., 2018. Financial networks and stress testing: Challenges and new research avenues for systemic risk analysis and financial stability implications. J. Fin. Stab. 35, 6–16. http://dx.doi.org/10.1016/j.jfs.2018.03.010.

Battiston, S., Puliga, M., Kaushik, R., Tasca, P., Caldarelli, G., 2012. DebtRank: Too central to fail? Financial networks, the FED and systemic risk. Sci. Rep. 2 (1), 541. http://dx.doi.org/10.1038/srep00541.

Basel Committee on Banking Supervision, 1988. Basel I: International convergence of capital measurement and capital standards.

Basel Committee on Banking Supervision, 2001. The internal ratings-based approach.

Basel Committee on Banking Supervision, 2004. Basel II: International convergence of capital measurement and capital standards.

Basel Committee on Banking Supervision, 2011. Basel III: A global regulatory framework for more resilient banks and banking systems revised version June 2011.

Basel Committee on Banking Supervision, 2015. Guidance on credit risk and accounting for expected credit losses.

Basel Committee on Banking Supervision, 2017. Guidance to banks on non-performing loans.

Basel Committee on Banking Supervision, 2021a. Climate-related financial risks – measurement methodologies.

Basel Committee on Banking Supervision, 2021b. Climate-related risk drivers and their transmission channels.

Basel Committee on Banking Supervision, 2022. Principles for the effective management and supervision of climate-related financial risks.

Bookstaber, R., Kenett, D.Y., et al., 2016. Looking deeper, seeing more: a multilayer map of the financial system. OFR Brief 16 (06).

Borio, C., Drehmann, M., Tsatsaronis, K., 2014. Stress-testing macro stress testing: Does it live up to expectations? J. Fin. Stab. 12, 3–15. http://dx.doi.org/10.1016/j.jfs.2013.06.001.

Borsos, A., Mero, B., 2020. Shock Propagation in the Banking System with Real Economy Feedback. Technical Report: MNB Working Papers.

Borsos, A., Stancsics, M., 2020. Unfolding the Hidden Structure of the Hungarian Multi-Layer Firm Network. Technical Report; MNB Occasional Papers.

Boss, M., Elsinger, H., Summer, M., Thurner, S., 2004. Network topology of the interbank market. Quant. Finance 4 (6), 677–684. http://dx.doi.org/10.1080/14697680400020325.

Brunetti, C., Dennis, B., Gates, D., Hancock, D., Ignell, D., Kiser, E.K., Kotta, G., Kovner, A., Rosen, R.J., Tabor, N.K., 2021. Climate Change and Financial Stability. FEDS Notes, Board of Governors of the Federal Reserve System, Washington, URL: https://doi.org/10.17016/2380-7172.2893.

Buncic, D., Melecky, M., 2013. Macroprudential stress testing of credit risk: A practical approach for policy makers. J. Fin. Stab. 9 (3), 347–370. http://dx.doi.org/10.1016/j.jfs.2012.11.003.

Burger, C., et al., 2022. Defaulting Alone: The Geography of Sme Owner Numbers and Credit Risk in Hungary. Technical Report, Magyar Nemzeti Bank (Central Bank of Hungary).

Capponi, A., Chen, P.-C., 2015. Systemic risk mitigation in financial networks. J. Econom. Dynam. Control 58, 152–166.

Carro, A., Stupariu, P., 2024. Uncertainty, non-linear contagion and the credit quality channel: an application to the Spanish interbank market. J. Fin. Stab. 101226.

Carvalho, V.M., Nirei, M., Saito, Y.U., Tahbaz-Salehi, A., 2021. Supply chain disruptions: Evidence from the great east Japan earthquake. Q. J. Econ. 136 (2), 1255–1321. http://dx.doi.org/10.1093/qje/qjaa044.

Cont, R., Moussa, A., Santos, E., 2010. Network structure and systemic risk in banking systems. http://dx.doi.org/10.2139/ssrn.1733528, SSRN.

Cont, R., Schaanning, E., 2017. Fire sales, indirect contagion and systemic stress testing. http://dx.doi.org/10.2139/ssrn.2541114, SSRN.

Cox, D.R., 1958. The regression analysis of binary sequences. J. R. Stat. Soc. Ser. B Stat. Methodol. 20 (2), 215–232. http://dx.doi.org/10.1111/j.2517-6161.1958.tb00292.x.

del Rio-Chanona, R.M., Mealy, P., Pichler, A., Lafond, F., Farmer, J.D., 2020. Supply and demand shocks in the COVID-19 pandemic: An industry and occupation perspective. Oxf. Rev. Econ. Policy 36 (Supplement_1), S94–S137.

Demir, B., Javorcik, B., Michalski, T.K., Ors, E., 2022. Financial constraints and propagation of shocks in production networks. Rev. Econ. Stat. 1–46. http://dx.doi.org/10.1162/rest_a_01162.

Diem, C., Borsos, A., Reisch, T., Kertész, J., Thurner, S., 2022a. Quantifying firm-level economic systemic risk from nation-wide supply networks. Sci. Rep. 12 (1), 7719. http://dx.doi.org/10.1038/s41598-022-11522-z.

Diem, C., Borsos, A., Reisch, T., Kertész, J., Thurner, S., 2022b. Supplementary information quantifying firm-level economic systemic risk from nation-wide supply networks. Sci. Rep. 12 (1), 7719. http://dx.doi.org/10.1038/s41598-022-11522-z.

Diem, C., Borsos, A., Reisch, T., Kertsz, J., Thurner, S., 2024a. Estimating the loss of economic predictability from aggregating firm-level production networks. PNAS Nexus 3 (3), pgae064. http://dx.doi.org/10.1093/pnasnexus/pgae064.

Diem, C., Pichler, A., Thurner, S., 2020. What is the minimal systemic risk in financial exposure networks? J. Econom. Dynam. Control 116, 103900. http://dx.doi.org/10.1016/j.jedc.2020.103900.

Diem, C., Schueller, W., Hinterplattner, M., Stangl, J., Conrady, B., Gerschberger, M., Thurner, S., 2024b. Supply network stress-testing of food security on the establishment-level. International Journal of Production Research 1–25. http://dx.doi.org/10.1080/00207543.2024.2416555.

EBA, 2020. EBA Report on Big Data and Advanced Analytics. European Banking Authority.

ECB, 2020. Our response to the coronavirus pandemic. URL: https://www.ecb.europa.eu/home/search/coronavirus/html/index.en.html.

ECB/ESRB, 2022. The Macroprudential Challenge of Climate Change. ESRB and ECB Report.

Elsinger, H., Lehar, A., Summer, M., 2006. Risk assessment for banking systems. Manage. Sci. 52 (9), 1301–1314. http://dx.doi.org/10.1287/mnsc.1060.0531.

Farmer, J.D., Kleinnijenhuis, A.M., Schuermann, T., Wetzer, T., 2022. Handbook of Financial Stress Testing. Cambridge University Press.

Feinstein, Z., Rudloff, B., Weber, S., 2017. Measures of systemic risk. SIAM J. Financial Math. 8 (1), 672–708. http://dx.doi.org/10.1137/16M1066087.

Gai, P., Kapadia, S., 2010. Contagion in financial networks. Proc. R. Soc. Lond. Ser. A 466 (2120), 2401–2423. http://dx.doi.org/10.1098/rspa.2009.0410.

Gao, J., Buldyrev, S.V., Havlin, S., Stanley, H.E., 2011. Robustness of a network of networks. Phys. Rev. Lett. 107, 195701. http://dx.doi.org/10.1103/PhysRevLett.107.195701.

Gauthier, C., Lehar, A., Souissi, M., 2012. Macroprudential capital requirements and systemic risk. J. Fin. Intermediation 21 (4), 594–618. http://dx.doi.org/10.1016/j.jfi.2012.01.005.

Glasserman, P., Young, H.P., 2016. Contagion in financial networks. J. Econ. Lit. 54 (3), 779–831.

Gorgijevska, A., Gorgieva-Trajkovska, O., 2019. Qualitative and quantitative analysis of creditworthiness of the companies. J. Econ. URL: https://js.ugd.edu.mk/index.php/JE/article/view/2736.

Gould, D.M., Kenett, D.Y., Panterov, G., 2021. Multi-dimensional economic connectivity: benefits, risks, and policy implications. Int. J. Finance Econ. 26 (4), 6110–6127.

Guth, M., Lipp, C., Puhr, C., Schneider, M., et al., 2020. Modeling the COVID-19 Effects on the Austrian Economy and Banking System, Vol. 40. Technical Report, Oesterreichische Nationalbank (Austrian Central Bank), pp. 63–86.

Hallegatte, S., 2008. An adaptive regional input-output model and its application to the assessment of the economic cost of Katrina. Risk Anal.: Int. J. 28 (3), 779–799. http://dx.doi.org/10.1111/j.1539-6924.2008.01046.x.

Hamilton, W.L., 2020. Graph representation learning. Synth. Lect. Artif. Intell. Mach. Learn. 14 (3), 1–159.

Herring, R.J., Schuermann, T., 2022. Objectives and challenges of stress testing. In: Handbook of Financial Stress Testing. Cambridge University Press, p. 9.

Huremovic, K., Jiménez, G., Moral-Benito, E., Vega-Redondo, F., Peydró, J.-L., et al., 2020. Production and financial networks in interplay: Crisis evidence from supplier-customer and credit registers.

IASB, 2014. International accounting standards (IFRS 9, financial instruments).

Inoue, H., Todo, Y., 2019. Firm-level propagation of shocks through supply-chain networks. Nat. Sustain. 2 (9), 841–847. http://dx.doi.org/10.1038/s41893-019-0351-x.

Klimek, P., Poledna, S., Doyne Farmer, J., Thurner, S., 2015. To bail-out or to bail-in? Answers from an agent-based model. J. Econom. Dynam. Control 50, 144–154. http://dx.doi.org/10.1016/j.jedc.2014.08.020.

Landaberry, V., Caccioli, F., Rodriguez-Martinez, A., Baron, A., Martinez-Jaramillo, S., Lluberas, R., 2021. The contribution of the intra-firm exposures network to systemic risk. Lat. Am. J. Cent. Bank. 2 (2), 100032. http://dx.doi.org/10.1016/j.latcb.2021.100032.

Levy-Carciente, S., Kenett, D.Y., Avakian, A., Stanley, H.E., Havlin, S., 2015. Dynamical macroprudential stress testing using network theory. J. Bank. Financ. 59, 164–181. http://dx.doi.org/10.1016/j.jbankfin.2015.05.008.

Martinez-Jaramillo, S., Carmona, C.U., Kenett, D.Y., 2019. Interconnectedness and financial stability. J. Risk Manag. Fin. Inst. 12 (2), 168–183.

McNeil, A.J., Frey, R., Embrechts, P., 2015. Quantitative Risk Ranagement: Concepts, Techniques and Tools. Princeton University Press.

Milstein, E., Wessel, D., 2021. What Did the Fed Do in Response to the COVID-19 Crisis. Vol. 17, Brookings Institution, URL: https://www.brookings.edu/research/fed-response-to-covid19/.

Moretto, A., Grassi, L., Caniato, F., Giorgino, M., Ronchi, S., 2019. Supply chain finance: From traditional to supply chain credit rating. J. Purch. Supply Manag. 25 (2), 197–217. http://dx.doi.org/10.1016/j.pursup.2018.06.004.

Pichler, A., Diem, C., Brintrup, A., Lafond, F., Magerman, G., Buiten, G., Choi, T.Y., Carvalho, V.M., Farmer, J.D., Thurner, S., 2023. Building an alliance to map global supply networks. Science 382 (6668), 270–272.

Pichler, A., Hurt, J., Reisch, T., Stangl, J., Thurner, S., 2024. Economic impacts of a drastic gas supply shock and short-term mitigation strategies. Journal of Economic Behavior & Organization 227, 106750.

Pichler, A., Pangallo, M., del Rio-Chanona, R.M., Lafond, F., Farmer, J.D., 2022b. Forecasting the propagation of pandemic shocks with a dynamic input-output model. J. Econom. Dynam. Control 144, 104527. http://dx.doi.org/10.1016/j.jedc.2022.104527.

Poledna, S., Hinteregger, A., Thurner, S., 2018. Identifying systemically important companies by using the credit network of an entire nation. Entropy 20 (10), http://dx.doi.org/10.3390/e20100792.

Poledna, S., Martínez-Jaramillo, S., Caccioli, F., Thurner, S., 2021. Quantification of systemic risk from overlapping portfolios in the financial system. J. Fin. Stab. 52, 100808. http://dx.doi.org/10.1016/j.jfs.2020.100808.

Poledna, S., Molina-Borboa, J.L., Martínez-Jaramillo, S., van der Leij, M., Thurner, S., 2015. The multi-layer network nature of systemic risk and its implications for the costs of financial crises. J. Fin. Stab. 20, 70–81. http://dx.doi.org/10.1016/j.jfs.2015.08.001.

Potter, S., Schuermann, T., 2020. Stressing the fed stress tests against COVID-19. http://dx.doi.org/10.2139/ssrn.3635187.

RegulationEU, 2013. Regulation (EU) no 575/2013 of the European parliament and of the council of 26 June 2013 on prudential requirements for credit institutions and investment firms and amending regulation (EU) no 648/2012.

Schivardi, F., Romano, G., 2020. Liquidity Crisis: Keeping Firms Afloat During Covid-19, vol. 18, VoxEU. org.

Shi, S., Tse, R., Luo, W., D'Addona, S., Pau, G., 2022. Machine learning-driven credit risk: a systemic review. Neural Comput. Appl. 1–13.

Silva, T.C., da Silva Alexandre, M., Tabak, B.M., 2018. Bank lending and systemic risk: A financial-real sector network approach with feedback. J. Fin. Stab. 38, 98–118. http://dx.doi.org/10.1016/j.jfs.2017.08.006.

Thurner, S., 2022. A complex systems perspective on macroprudential regulation. In: Farmer, J.D., et al. (Eds.), Handbook of Financial Stress Testing. Vol. 5, Cambridge University Press, pp. 593–634, URL: https://www.cambridge.org/core/books/abs/handbook-of-financial-stress-testing/complex-systems-perspective-on-macroprudential-regulation/4317CB6A85321EBCA52F4DBC64E58330.

Thurner, S., Poledna, S., 2013. DebtRank-transparency: Controlling systemic risk in financial networks. Sci. Rep. 3, 1888. http://dx.doi.org/10.1038/srep01888.

Vazquez, F., Tabak, B.M., Souto, M., 2012. A macro stress test model of credit risk for the Brazilian banking sector. J. Fin. Stab. 8 (2), 69–83. http://dx.doi.org/10.1016/j.jfs.2011.05.002.

Willner, S.N., Otto, C., Levermann, A., 2018. Global economic response to river floods. Nature Clim. Change 8 (7), 594–598. http://dx.doi.org/10.1038/s41558-018-0173-2.